

claude-windows / f20c94c2-0ea3-45f5-9aea-07136e11e4d5

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\f20c94c2-0ea3-45f5-9aea-07136e11e4d5.jsonl`
- SHA-256: `bb6b8d2bdb7f855c406e3310ab3bf18b87ef3724813332f4bc0d0f5eb2cc7783`
- Source modified: `2026-06-22T09:03:01+00:00`
- Imported at: `2026-07-05T16:48:19+00:00`
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`
- session_id: `f20c94c2-0ea3-45f5-9aea-07136e11e4d5`
```

Transcript

1. user (2026-06-13T19:49:24.869Z)

Improve `run.log` readability for the owner's manual-QA workflow: he runs the tool on a video, then eyeballs the rally CSV against the log to verify quality. Two changes, design-then-implement, ONE PR off master.

FIRST ? branching safety (this checkout is SHARED by concurrent sessions): `git fetch origin` then `git checkout -b <branch> origin/master` (branch from the REMOTE, not the current branch). Re-verify `git branch --show-current` + `git log --oneline -1` right before committing; before the PR, `git log --oneline origin/master..HEAD` must show ONLY your commits. Stage EXPLICIT paths (never `git add -A`). Don't merge without owner.

INVESTIGATE (don't assume): find where logging is configured (`grep -rn "run.log\|basicConfig\|FileHandler\|getLogger" backend/ run\_\*.py`), how the Gemini path logs (it uses `google-genai` ? `httpx`/`grpc`/`google.auth` ? those emit the noisy per-request HTTP lines), where per-rally lines are already logged (`[rally]` / `[<label> rally]` in `backend/pipeline/segmenters/{trajectory\_hybrid,yolo\_hybrid}.py`), and where the rally CSV is written (`backend/eval/export\_wasb\_segments.py::write\_segments\_csv` and/or the reporting layer `backend/reporting/`). Read `docs/CODE\_MAP.md` for the map.

CHANGE 1 ? suppress HTTP chatter: raise the noisy third-party loggers (`httpx`, `httpcore`, `urllib3`, `google\_genai`/`google.genai`, `google.auth`, `grpc`) to WARNING by default, behind a config knob (e.g. `logging.verbose\_http: false`) so debugging can re-enable them. Do it in the central logging setup, not scattered. Don't suppress our own `backend.\*` logs.

CHANGE 2 ? aggregate rally-times summary: at the END of a run, log a compact, scannable block ? total rallies, total highlight duration, and a per-rally table `# start(HH:MM:SS) end dur(s) shot\_count` ? sourced from the produced play_segments (the same data that goes to the rally CSV), so the log and CSV line up for manual verification. Make it a small pure helper (format a list of segments ? text) so it's unit-testable offline, then call it from the run path. Keep timestamps in HH:MM:SS to match the existing `\_fmt\_ts`.

Deliverable: a short plan in the PR description + the implementation + tests (`python run\_all\_tests.py` green, ruff + mypy clean). Keep it minimal and config-gated; default behavior = quieter log + the summary block. Don't touch unrelated areas.

2. assistant (2026-06-13T19:49:35.196Z)

I'll start by setting up the branch safely and investigating the codebase in parallel.

3. user (2026-06-13T19:49:41.410Z)

35a50f3 docs(readme): index QUALITY_ITERATIONS + mark EXPERIMENT_HARNESS P1 SHIPPED (#118)
cbf2236 feat(segmenters): FusionSegmenter ? shuttle + net-crossing + person-cue fusion (P2, default-OFF) (#117)
df7d5df refactor(detectors): extract the person cue into PersonDetector (fusion P1) (#115)

4. user (2026-06-13T19:49:41.591Z)

```
run_regression.py:53: logging.basicConfig(level=logging.INFO, format="%(levelname)s
%(message)s")
run_eval.py:75: logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")
run_phase3_eval.py:41: logger = logging.getLogger(__name__)
run_phase3_eval.py:76: logging.basicConfig(level=logging.INFO, format="%(levelname)s
%(message)s")
run_process_and_stitch.py:8:# Configure root logger ? all backend modules use
logging.getLogger(__name__)
run_process_and_stitch.py:9:logging.basicConfig(
scratch\run_gx010125.py:18:logging.basicConfig(
```

5. user (2026-06-13T19:49:41.610Z)

```
backend\main.py:15:logging.basicConfig(
backend\main.py:20:logger = logging.getLogger(__name__)
backend\config\loader.py:20:logger = logging.getLogger(__name__)
backend\providers\gemini.py:11:logger = logging.getLogger(__name__)
backend\infrastructure\database.py:7:logger = logging.getLogger(__name__)
backend\utils\video.py:5:logger = logging.getLogger(__name__)
backend\annotations\automated_provider.py:30:logger = logging.getLogger(__name__)
backend\eval\gt_loader.py:26:logger = logging.getLogger(__name__)
backend\pipeline\stitcher\compiler.py:8:logger = logging.getLogger(__name__)
backend\eval\experiment.py:55:logger = logging.getLogger(__name__)
backend\pipeline\segmenters\gemini.py:13:logger = logging.getLogger(__name__)
backend\pipeline\detectors\person_detector.py:33:logger = logging.getLogger(__name__)
backend\pipeline\segmenters\motion.py:9:logger = logging.getLogger(__name__)
backend\pipeline\detectors\base.py:75:logger = logging.getLogger(__name__)
backend\eval\harness.py:25:logger = logging.getLogger(__name__)
backend\eval\regression.py:29:logger = logging.getLogger(__name__)
backend\pipeline\segmenters\trajectory_hybrid.py:34:logger = logging.getLogger(__name__)
backend\pipeline\detectors\tracknet_runner.py:27:logger = logging.getLogger(__name__)
backend\pipeline\segmenters\yolo_hybrid.py:30:logger = logging.getLogger(__name__)
backend\eval\annotations.py:19:logger = logging.getLogger(__name__)
backend\pipeline\detectors\wasb_runner.py:24:logger = logging.getLogger(__name__)
backend\pipeline\detectors\wasb_infer.py:48:logger = logging.getLogger(__name__)
```

6. assistant (2026-06-13T19:49:45.323Z)

Let me look at where `run.log` is configured and read the central logging setup files.

7. user (2026-06-13T19:49:46.927Z)

```
tools\cleanup_caches.py:58: if low.endswith((".log", ".sh", ".txt", ".json", ".md")):
```

8. user (2026-06-13T19:49:47.147Z)

```
1 import os
2 import json
3 import logging
4 from dotenv import load_dotenv
5
6 load_dotenv()
7
8 # Configure root logger ? all backend modules use logging.getLogger(__name__)
9 logging.basicConfig(
10     level=logging.INFO,
11     format="%(asctime)s [%(name)s] %(levelname)s: %(message)s",
12     datefmt="%H:%M:%S",
13 )
14
15 from backend.infrastructure.database import Database
16 from backend.config import load_config
```

```

17     from backend.utils.hashing import compute_video_id
18     from backend.pipeline.segmenters import segmenter_registry
19     from backend.pipeline.stitcher.compiler import VideoCompiler
20
21     def main():
22         video_path = r"C:\Users\avidu\Videos\Testing\KushagraYashAviFirstVideo.mp4"
23         output_dir = r"C:\Users\avidu\Videos\Testing"
24         output_filename = "Final_2_OutputOfKushagraYashAviFirstVideo.mp4"
25
26         print("Loading config...")
27         cfg = load_config() # typed AppConfig ? single source of defaults
28
29         print("Verifying files exist...")
30         if not os.path.exists(video_path):
31             print(f"Error: Video file not found at {video_path}")
32             return
33
34         os.makedirs(cfg.output.output_dir, exist_ok=True)
35         db_path = os.path.join(cfg.output.output_dir, cfg.output.db_name)
36         db = Database(db_path)
37
38         # Calculate MD5 of the video to uniquely identify it
39         import time
40         print("Calculating video MD5 (this might take a few seconds on large files)...",
41 flush=True)
42         t0 = time.time()
43         video_id = compute_video_id(video_path)
44         print(f"Calculated MD5: {video_id} (took {time.time()-t0:.1f}s)", flush=True)
45
46         segments = []
47         cached_video = db.get_video(video_id)
48         cache_hit = False
49
50         if cached_video and cached_video.get("status") == "processed":
51             segments = db.query_play_segments(video_id, {})
52             if len(segments) > 0:
53                 print(f"\n[CACHE] Found cached indexed video in database (MD5: {video_id})
54 with {len(segments)} parsed play segments.", flush=True)
55                 choice = input("Would you like to (1) Run stitching on the cached segments,
56 or (2) Reprocess the video with Gemini from scratch? [1/2]: ").strip()
57                 if choice == '1':
58                     print("[CACHE] Using cached play segments. Skipping segmenter API
59 call.", flush=True)
60                     cache_hit = True
61                 else:
62                     print("[CACHE] Ignoring cache...", flush=True)
63                 else:
64                     print("[CACHE] Video exists in DB but contains 0 play segments. Treating as
65 cache miss.", flush=True)
66
67         if not cache_hit:
68             print("[CACHE] Initializing clean indexing...", flush=True)
69             db.clear_video_data(video_id)
70
71         print("Bypassing OpenCV validations for performance on massive files...",
72 flush=True)
73         db.add_video(video_id, video_path, "processed", [])
74
75         print("Running segmenter & indexing...", flush=True)
76         seg_name = cfg.indexing.default_segmenter
77         segmenter_cls = segmenter_registry.get(seg_name)
78         if not segmenter_cls:
79             print(f"[FAIL] Segmenter '{seg_name}' not found.")
80             return
81         print(f"Using segmenter: {seg_name}", flush=True)

```

```

76         segmenter = segmenter_cls(db, cfg)
77         segments = segmenter.process_video(video_id, video_path)
78         print(f"Successfully indexed {len(segments)} play segments.", flush=True)
79
80     if not segments:
81         print("No play segments detected. Cannot compile highlights.")
82         return
83
84
85     end_padding = cfg.stitching.end_padding
86     begin_padding = cfg.stitching.begin_padding
87     min_shot_count = cfg.stitching.min_shot_count
88
89     if cache_hit:
90         print("\n--- Stitching Customizations ---")
91         try:
92             bpad_input = input(f"Enter begin_padding in seconds (default
{begin_padding}): ").strip()
93             if bpad_input:
94                 begin_padding = float(bpad_input)
95
96             pad_input = input(f"Enter end_padding in seconds (default {end_padding}):
").strip()
97             if pad_input:
98                 end_padding = float(pad_input)
99
100             shot_input = input(f"Enter min_shot_count to filter by (default
{min_shot_count}): ").strip()
101             if shot_input:
102                 min_shot_count = int(shot_input)
103         except ValueError:
104             print("[WARNING] Invalid input detected, falling back to defaults from
config.json.")
105
106         print(f"\nCompiling highlights (with begin_padding={begin_padding}s,
end_padding={end_padding}s, min_shot_count={min_shot_count})...", flush=True)
107
108         # Filter segments where shot_count >= min_shot_count
109         time_pairs = []
110         skipped_count = 0
111         for s in segments:
112             meta = s.get("metadata", {})
113             if isinstance(meta, str):
114                 try:
115                     meta = json.loads(meta)
116                 except Exception:
117                     meta = {}
118             shot_count = meta.get("shot_count", 0)
119             if shot_count >= min_shot_count:
120                 time_pairs.append((s["start_time"], s["end_time"]))
121             else:
122                 skipped_count += 1
123
124         if skipped_count > 0:
125             print(f"[FILTER] Excluded {skipped_count} play segments with fewer than
{min_shot_count} shots.", flush=True)
126         print(f"[FILTER] Retaining {len(time_pairs)} play segments for stitching.",
flush=True)
127
128         if not time_pairs:
129             print("No play segments passed the filtering criteria. Cannot compile
highlights.")
130             return
131
132         compiler = VideoCompiler(output_dir, end_padding=end_padding,

```

```

begin_padding=begin_padding,
133                                     watermark=cfg.stitching.watermark)
134     try:
135         out_path = compiler.compile_highlights(video_path, time_pairs, output_filename)
136         print(f"Stitched highlight video saved to: {out_path}", flush=True)
137     except Exception as e:
138         print(f"Error during stitching: {e}", flush=True)
139
140 if __name__ == "__main__":
141     main()
142

```

9. assistant (2026-06-13T19:49:59.524Z)

`run.log` isn't a literal in code ? it's a stdout/stderr redirect capturing the `logging` output configured by `basicConfig`. Let me read the central pieces: `main.py` logging, the config structure, the segmenter rally-logging, and the CSV writer.

10. user (2026-06-13T19:50:00.483Z)

```

1     import os
2     import sys
3     import argparse
4     import logging
5     import threading
6     import traceback
7     import uuid
8     from concurrent.futures import ThreadPoolExecutor
9     import uvicorn
10    from dotenv import load_dotenv
11
12    load_dotenv()
13
14    # Centralized logging setup for the indexer (hot paths use logger, CLI output uses print
for direct user-facing messages).
15    logging.basicConfig(
16        level=logging.INFO,
17        format='%(asctime)s [%(levelname)s] %(name)s: %(message)s',
18        datefmt='%H:%M:%S'
19    )
20    logger = logging.getLogger(__name__)
21    from fastapi import FastAPI, HTTPException, Request
22    from fastapi.middleware.cors import CORSMiddleware
23    from pydantic import BaseModel
24    from typing import List, Dict, Any, Optional
25
26    from fastapi.staticfiles import StaticFiles
27    from fastapi.responses import FileResponse, HTMLResponse, JSONResponse
28    from backend.pipeline.validators.base import registry
29    from backend.infrastructure.database import Database
30    from backend.pipeline.segmenters.base import segmenter_registry

```

11. user (2026-06-13T19:50:01.394Z)

```

1     """Export tuned WASB rally segments for manual eyeball verification.
2
3     After calibration picks a tuned threshold config, this turns a cached WASB
4     trajectory into the actual rally [start,end] segments that config produces, so the
5     user can answer "are the rallies actually better?" ? by scrubbing the video to each
6     segment, by diffing tuned-vs-golden quantitatively, or by cutting clips.
7
8     Reuses (no new windowing/metrics/cutting logic):
9     - calibrate_wasb.make_predict_fn -> trajectory -> segments for a given config
10    - eval.metrics.match_intervals/score -> tuned-vs-golden IoU comparison

```

```

11     - tools.rally_slicer.slice_rallies    -> optional clip cutting (the exported CSV is
12       written in the golden/slicer schema, so it feeds straight back in)
13
14     Run (uses the tuned config from the first calibration by default):
15     python -m backend.eval.export_wasb_segments --trajectory
~/clips/sample_long_traj.csv \
16         --labels "Annotation Setup/golden_labels_badminton.csv" --fps 59.94 --frame-
width 1920
17         # add --video <mp4> --slice to also cut one padded clip per tuned rally
18     """
19     from __future__ import annotations
20
21     import csv
22     import os
23     import os.path as osp
24     import argparse
25     from typing import List, Tuple, Dict, Optional
26
27     from backend.eval.calibrate_wasb import make_predict_fn, load_golden_gts, BASELINE
28     from backend.eval.metrics import match_intervals, score
29     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
30     from backend.pipeline.detectors import rally_gate
31
32     Interval = Tuple[float, float]
33     Config = Tuple[float, float, float]
34
35     # The tuned config from the first per-video calibration (velocity, merge_gap, min_dur);
36     # F1 0.588 (baseline) -> 0.746. Override on the CLI to export a different operating
point.
37     TUNED: Config = (0.05, 1.0, 1.0)
38
39     SEG_FIELDS = ["rally_number", "start_time", "end_time", "duration", "start_mmss",
"end_mmss"]
40     COMP_FIELDS = ["type", "ref", "start_time", "end_time", "mmss", "status",
41                   "iou", "pred_start", "pred_end", "start_err_s", "end_err_s"]
42
43
44     def seconds_to_mmss(t: float) -> str:
45         """Human-scrubbable timestamp, e.g. 106.031 -> '1:46.031'."""
46         t = max(0.0, float(t))
47         m = int(t // 60)
48         return f"{m}:{t - m * 60:06.3f}"
49
50
51     def segments_to_rows(segments: List[Interval]) -> List[Dict[str, str]]:
52         """Number segments 1..N in time order; emit the golden/slicer CSV schema."""
53         rows = []
54         for i, (s, e) in enumerate(segments, start=1):
55             rows.append({
56                 "rally_number": str(i),
57                 "start_time": f"{s:.3f}",
58                 "end_time": f"{e:.3f}",
59                 "duration": f"{e - s:.3f}",
60                 "start_mmss": seconds_to_mmss(s),
61                 "end_mmss": seconds_to_mmss(e),
62             })
63         return rows
64
65
66     def _ensure_parent(path: str) -> None:
67         os.makedirs(osp.dirname(osp.abspath(path)) or ".", exist_ok=True)
68
69
70     def write_segments_csv(path: str, segments: List[Interval]) -> str:
71         """Write segments in the golden schema (rally_number/start_time/end_time + mm:ss).

```

```

72
73     The schema is what tools.rally_slicer.load_rally_labels consumes, so the file
74     doubles as input to clip cutting.
75     """
76     _ensure_parent(path)
77     with open(path, "w", newline="", encoding="utf-8") as f:
78         w = csv.DictWriter(f, fieldnames=SEG_FIELDS)
79         w.writeheader()
80         w.writerows(segments_to_rows(segments))
81     return path
82
83
84     def build_comparison(preds: List[Interval], gts: List[Interval],
85                         iou_threshold: float = 0.5) -> dict:
86         """Golden-centric comparison: per golden rally, did a tuned segment match (IoU),
87         plus the boundary errors and any extra (false-positive) tuned segments."""
88         mr = match_intervals(preds, gts, iou_threshold)
89         sc = score(preds, gts, iou_threshold, match_result=mr)
90         by_gt = {m.gt_index: m for m in mr.matches}
91
92         rows: List[Dict[str, str]] = []
93         for gi, (gs, ge) in enumerate(gts, start=0):
94             m = by_gt.get(gi)
95             if m is not None:
96                 ps, pe = preds[m.pred_index]
97                 rows.append({
98                     "type": "golden", "ref": str(gi + 1),
99                     "start_time": f"{gs:.3f}", "end_time": f"{ge:.3f}",
100                     "mmss": f"{seconds_to_mmss(gs)}-{seconds_to_mmss(ge)}",
101                     "status": "matched", "iou": f"{m.iou:.3f}",
102                     "pred_start": f"{ps:.3f}", "pred_end": f"{pe:.3f}",
103                     "start_err_s": f"{ps - gs:+.3f}", "end_err_s": f"{pe - ge:+.3f}",
104                 })
105             else:
106                 rows.append({
107                     "type": "golden", "ref": str(gi + 1),
108                     "start_time": f"{gs:.3f}", "end_time": f"{ge:.3f}",
109                     "mmss": f"{seconds_to_mmss(gs)}-{seconds_to_mmss(ge)}",
110                     "status": "MISS", "iou": "0.000",
111                     "pred_start": "", "pred_end": "", "start_err_s": "", "end_err_s": "",
112                 })
113         for pi in mr.unmatched_pred_indices:           # extra tuned segments = false
positives
114             ps, pe = preds[pi]
115             rows.append({
116                 "type": "pred_extra", "ref": str(pi + 1),
117                 "start_time": f"{ps:.3f}", "end_time": f"{pe:.3f}",
118                 "mmss": f"{seconds_to_mmss(ps)}-{seconds_to_mmss(pe)}",
119                 "status": "false_positive", "iou": "",
120                 "pred_start": "", "pred_end": "", "start_err_s": "", "end_err_s": "",
121             })
122         return {"score": sc, "rows": rows}
123
124
125     def write_comparison_csv(path: str, comparison: dict) -> str:
126         _ensure_parent(path)
127         with open(path, "w", newline="", encoding="utf-8") as f:
128             w = csv.DictWriter(f, fieldnames=COMP_FIELDS)
129             w.writeheader()
130             w.writerows(comparison["rows"])
131         return path
132
133
134     def _with_suffix(path: str, suffix: str) -> str:
135         root, ext = osp.splitext(path)

```

```

136         return root + suffix + ext
137
138
139     def run(trajjectory_csv: str, fps: float, frame_width: float, cfg: Config = TUNED,
140            out_csv: str = "output/wasb_tuned_segments.csv", labels_csv: Optional[str] =
None,
141            iou: float = 0.5, baseline: bool = False,
142            comparison_out: Optional[str] = None, min_crossings: int = 0) -> dict:
143         predict_fn = make_predict_fn(trajjectory_csv, fps, frame_width)
144         # Gate A: drop between-point false fires (single service-feed tosses) by requiring
145         # the shuttle to cross the net >= min_crossings times within the window.
146         points = TrackNetRunner.parse_trajectory_csv(trajjectory_csv) if min_crossings > 0
147     else None
148
149     def _gate(segs: List[Interval], label: str) -> List[Interval]:
150         if not points:
151             return segs
152         kept = rally_gate.apply_gate(points, segs, fps, min_crossings=min_crossings)
153         print(f"[gate] {label}: {len(segs)} -> {len(kept)} segments "
154               f"(removed {len(segs) - len(kept)} with <{min_crossings} net-crossings)")
155         return kept
156
157     tuned_segs = _gate(predict_fn(cfg), "tuned")
158     write_segments_csv(out_csv, tuned_segs)
159     total = sum(e - s for s, e in tuned_segs)
160     print("=== WASB tuned-segment export ===")
161     print(f"tuned cfg {cfg}: {len(tuned_segs)} rally segments, {total:.1f}s of play ->
{out_csv}")
162
163     result: dict = {"tuned_cfg": cfg, "num_tuned": len(tuned_segs), "out_csv": out_csv}
164
165     if baseline:
166         base_segs = _gate(predict_fn(BASELINE), "baseline")
167         base_out = _with_suffix(out_csv, "_baseline")
168         write_segments_csv(base_out, base_segs)
169         print(f"baseline cfg {BASELINE}: {len(base_segs)} segments -> {base_out}")
170         result["num_baseline"] = len(base_segs)
171         result["baseline_csv"] = base_out
172
173     if labels_csv:
174         gts = load_golden_gts(labels_csv)
175         if not gts:
176             raise SystemExit(f"no usable golden rallies in {labels_csv}")
177         comp = build_comparison(tuned_segs, gts, iou)
178         sc = comp["score"]
179         comp_out = comparison_out or _with_suffix(out_csv, "_vs_golden")
180         write_comparison_csv(comp_out, comp)
181         print(f"\n--- tuned vs golden ({len(gts)} rallies, IoU>={iou}) ---")
182         print(f"matched (TP) {sc.true_positives} | misses (FN) {sc.false_negatives} | "
183               f"extra (FP) {sc.false_positives}")
184         print(f"precision {sc.precision:.3f} | recall {sc.recall:.3f} | f1 {sc.f1:.3f} |
"
185               f"meanIoU {sc.mean_iou:.3f}")
186         print(f"mean boundary error: start {sc.mean_start_error:.2f}s | end
{sc.mean_end_error:.2f}s")
187         print(f"per-rally detail -> {comp_out}")
188         result["comparison"] = sc.as_dict()
189         result["comparison_csv"] = comp_out
190     return result
191
192     def _maybe_slice(video: str, segments_csv: str, clips_dir: str) -> None:
193         """Cut one padded clip per exported rally (reuses the rally slicer)."""
194         from backend.tools.rally_slicer import slice_rallies
195         written = slice_rallies(video, segments_csv, ["all"], clips_dir)
196         print(f"\ncut {len(written)} clip(s) -> {clips_dir}")

```



```

196
197
198     def main() -> None:
199         ap = argparse.ArgumentParser(description="Export tuned WASB rally segments for
200 eyeballing.")
201         ap.add_argument("--trajectory", required=True, help="WASB trajectory CSV
(Frame,Visibility,X,Y)")
202         ap.add_argument("--fps", type=float, default=59.94)
203         ap.add_argument("--frame-width", type=float, default=1920.0)
204         ap.add_argument("--velocity-thresh", type=float, default=TUNED[0])
205         ap.add_argument("--merge-gap", type=float, default=TUNED[1])
206         ap.add_argument("--min-duration", type=float, default=TUNED[2])
207         ap.add_argument("--labels", default=None, help="golden labels CSV -> tuned-vs-golden
comparison")
208         ap.add_argument("--iou", type=float, default=0.5)
209         ap.add_argument("--out", default="output/wasb_tuned_segments.csv")
210         ap.add_argument("--comparison-out", default=None)
211         ap.add_argument("--baseline", action="store_true", help="also export baseline-config
segments")
212         ap.add_argument("--min-crossings", type=int, default=0,
213                         help="gate A: drop windows whose shuttle crosses the net < N times "
214                         "(kills single-toss service-feed false fires; 0=off, 2
recommended)")
215         ap.add_argument("--video", default=None, help="source video (with --slice) to cut
clips from")
216         ap.add_argument("--slice", action="store_true", help="cut one padded clip per tuned
rally")
217         ap.add_argument("--clips-dir", default="output/wasb_tuned_clips")
218         args = ap.parse_args()
219
220         cfg = (args.velocity_thresh, args.merge_gap, args.min_duration)
221         res = run(args.trajectory, args.fps, args.frame_width, cfg=cfg, out_csv=args.out,
222                 labels_csv=args.labels, iou=args.iou, baseline=args.baseline,
223                 comparison_out=args.comparison_out, min_crossings=args.min_crossings)
224         if args.slice:
225             if not args.video:
226                 raise SystemExit("--slice requires --video")
227             _maybe_slice(args.video, res["out_csv"], args.clips_dir)
228
229     if __name__ == "__main__":
230         main()
231

```

12. user (2026-06-13T19:50:01.470Z)

```

backend\config\loader.py
backend\config\__init__.py
backend\config\models.py

```

13. assistant (2026-06-13T19:50:11.467Z)

The WASB export CSV is the eval path. The owner's manual-QA runs `run\_process\_and\_stitch.py` over play_segments. Let me find `\_fmt\_ts`, the reporting/CSV layer, the rally-logging lines, and the config model.

14. user (2026-06-13T19:50:13.009Z)

```

backend\config\models.py:193:     min_shot_count: int = 1
backend\main.py:502:     # The configured shot-count floor (stitching.min_shot_count) applies on
the API
backend\main.py:503:     # path too ? parity with run_process_and_stitch.py. Segments WITHOUT
shot_count
backend\main.py:510:     sc = meta.get("shot_count") if isinstance(meta, dict) else None

```

```

backend\main.py:512:         if sc is not None and int(sc) < stitching.min_shot_count:
backend\main.py:518:         logger.info(f"[compile]
stitching.min_shot_count={stitching.min_shot_count}: dropped "
backend\main.py:523:
f"stitching.min_shot_count={stitching.min_shot_count}.")
backend\tools\export.py:76:     """Return a CSV string
(start,end,duration,shot_count,detector,...) .
backend\tools\export.py:84:
lines.append("start_time,end_time,duration,shot_count,shot_count_confidence,detector,id")
backend\tools\export.py:87:         shot = md.get("shot_count", "")
backend\tools\export.py:88:         conf = md.get("shot_count_confidence", "")
backend\sports\badminton.py:35:Confidence Scoring: Because the shuttlecock moves at extreme
velocities that may fall between sampled frames, provide a shot_count_confidence metric (Low,
Medium, High). Base your estimate on player biomechanics, positioning, and racket swings if the
shuttle itself is blurred.
backend\sports\badminton.py:46:"shot_count_confidence": "High"
backend\sports\badminton.py:52:"shot_count_confidence": "Medium"
backend\tools\rally_slicer.py:72:         labels[rally] = (start, end)
backend\tools\rally_slicer.py:93:         start, end = labels[rally]
backend\eval\harness.py:85:def _fmt_ts(seconds: float) -> str:
backend\eval\harness.py:134:         lines.append(f" -
{_fmt_ts(iv[0])}?{_fmt_ts(iv[1])}")
backend\eval\harness.py:137:         lines.append(f" -
{_fmt_ts(iv[0])}?{_fmt_ts(iv[1])}")
backend\pipeline\segmenters\gemini.py:428:         # Extract shot_count from Gemini or
compute estimate if not present
backend\pipeline\segmenters\gemini.py:429:         shot_count =
segment.get("shuttle_exchange_count")
backend\pipeline\segmenters\gemini.py:430:         if shot_count is None:
backend\pipeline\segmenters\gemini.py:431:             shot_count = max(3, int(dur / 0.8)) #
1 shot every 0.8 seconds as fallback estimate
backend\pipeline\segmenters\gemini.py:434:             shot_count = int(shot_count)
backend\pipeline\segmenters\gemini.py:436:             shot_count = max(3, int(dur /
0.8))
backend\pipeline\segmenters\gemini.py:440:             "shot_count": shot_count,
backend\pipeline\segmenters\gemini.py:441:             "shot_count_confidence":
segment.get("shot_count_confidence", "Unknown"),
backend\pipeline\segmenters\motion.py:168:             "shot_count": random.randint(4, 25),
backend\pipeline\segmenters\trajectory_hybrid.py:37:def _fmt_ts(seconds: float) -> str:
backend\pipeline\segmenters\trajectory_hybrid.py:67:     def _shot_count_from(segment: Dict[str,
Any], duration: float) -> int:
backend\pipeline\segmenters\trajectory_hybrid.py:73:         shot_count =
segment.get("shuttle_exchange_count")
backend\pipeline\segmenters\trajectory_hybrid.py:74:         if shot_count is None:
backend\pipeline\segmenters\trajectory_hybrid.py:77:             return int(shot_count)
backend\pipeline\segmenters\trajectory_hybrid.py:196:             logger.info(f"[{label}]
rally] {_fmt_ts(cw['start'])}-{_fmt_ts(cw['end'])}"
backend\pipeline\segmenters\trajectory_hybrid.py:276:             "shot_count":
self._shot_count_from(segment, dur),
backend\pipeline\segmenters\trajectory_hybrid.py:277:
"shot_count_confidence": segment.get("shot_count_confidence", "Unknown"),
backend\pipeline\segmenters\yolo_hybrid.py:33:def _fmt_ts(seconds: float) -> str:
backend\pipeline\segmenters\yolo_hybrid.py:241:         f"[rally]
{_fmt_ts(w['start'])}-{_fmt_ts(w['end'])}"
backend\pipeline\segmenters\yolo_hybrid.py:333:         shot_count =
segment.get("shuttle_exchange_count")
backend\pipeline\segmenters\yolo_hybrid.py:334:         if shot_count is None:
backend\pipeline\segmenters\yolo_hybrid.py:335:             shot_count = max(3, int(dur /
0.8))
backend\pipeline\segmenters\yolo_hybrid.py:338:             shot_count =
int(shot_count)
backend\pipeline\segmenters\yolo_hybrid.py:340:             shot_count = max(3,
int(dur / 0.8))
backend\pipeline\segmenters\yolo_hybrid.py:343:             "shot_count": shot_count,
backend\pipeline\segmenters\yolo_hybrid.py:344:             "shot_count_confidence":

```

```
segment.get("shot_count_confidence", "Unknown"),
```

15. user (2026-06-13T19:50:13.800Z)

```
1      """Pydantic models for the application configuration.
2
3      This module defines the canonical, validated, typed configuration for the
4      Badminton Highlight Indexer. Built-in defaults live here (single source of truth).
5
6      The models support graceful migration from old config.json files via extra="allow".
7      """
8
9      from __future__ import annotations
10
11     from typing import Any, List, Literal, Optional
12
13     from pydantic import BaseModel, Field, field_validator, model_validator
14
15     Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]
16
17
18     class ValidationRelevanceConfig(BaseModel):
19         court_green_lower: list[int] = Field(default=[35, 40, 40])
20         court_green_upper: list[int] = Field(default=[85, 255, 255])
21         court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)
22
23
24     class ValidationConfig(BaseModel):
25         min_resolution_width: int = 1280
26         min_resolution_height: int = 720
27         min_fps: float = 29.9
28         min_blur_threshold: float = 20.0
29         max_scene_cuts_per_minute: float = 0.5
30         relevance: ValidationRelevanceConfig =
31         Field(default_factory=ValidationRelevanceConfig)
32
33     class TrackNetConfig(BaseModel):
34         wsl_distro: str = "Ubuntu"
35         repo_dir: str = "~/models/TrackNetV4"
36         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
37         conda_env: str = "TrackNetV4"
38         weights_path: str = ""
39         queue_length: int = 5
40         stage_in_wsl: bool = True
41         wsl_stage_dir: str = "~/clips"
42         timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
43         velocity_threshold: float = 0.02
44         # Cache hygiene: after a SUCCESSFUL run the staged video copy (and, for WASB, the
45         # decoded frame cache) are deleted automatically ? they are pure intermediates,
46         # regenerable from the source, and the dominant disk consumer (~GBs/video). Set
47         # true only for debugging. On FAILURE they are always kept so a re-run can resume.
48         keep_frames: bool = False
49         # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB.
50         # 0 = disabled.
51         min_free_gb: float = 0.0
52
53
54     class WasbIndexConfig(BaseModel):
55         """Typed config for the live WASB shuttle substrate (indexing.wasb).
56
57         Mirrors backend/pipeline/detectors/wasb_runner.py::WasbConfig.from_indexing_cfg so
58         these knobs actually take effect ? previously this whole block was silently dropped
59         because nested config sections defaulted to extra='ignore'.
60         """
```

```

61     wsl_distro: str = "Ubuntu"
62     repo_dir: str = "~/models/WASB-SBDT"
63     conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
64     conda_env: str = "wasb"
65     weights_path: str = "~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
66     sport: str = "badminton"
67     wsl_stage_dir: str = "~/clips"
68     timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
69     velocity_threshold: float = 0.02
70     # Cache hygiene: after a SUCCESSFUL run the staged video copy + decoded frame cache
71     # (the ~GBs/video disk hog) are deleted automatically ? pure intermediates,
regenerable
72     # from the source. Set true only for debugging. On FAILURE they are kept so a re-run
resumes.
73     keep_frames: bool = False
74     # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB. 0 =
disabled.
75     min_free_gb: float = 0.0
76
77
78     class FusionConfig(BaseModel):
79         """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN
P2).
80
81         Every default reproduces control behaviour: the fusion segmenter persists the
82         shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']``
is
83         true ? the person-cue features, but it does NOT gate the served handoff unless a
84         threshold is raised (``min_crossings``/``min_players`` default 0 = OFF). The court
mask
85         is optional: ``court_polygon=None`` ? Gate B counts every detection (player-count-
only);
86         supplied ? off-court persons (spectators / adjacent courts) are excluded.
87         """
88         # Which trajectory substrate seeds the windows (shuttle stays primary).
89         substrate: Literal["wasb", "tracknet"] = "wasb"
90         # Windowing velocity threshold for the fusion family (the engine reads
91         # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
92         # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
93         velocity_threshold: float = 0.02
94         # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
95         # enabled ? so the default path never imports torch / touches the GPU.
96         cue_flags: dict[str, bool] = Field(default_factory=dict)
97         # Served Gate A: drop windows whose shuttle net-crossings < this from the AI
handoff.
98         # 0 = OFF (no gating; persisted candidates are always the full superset for offline
re-scoring).
99         min_crossings: int = Field(default=0, ge=0)
100         # Served Gate B: when the person cue is on, drop windows with median in-court
players
101         # < this. 0 = OFF.
102         min_players: int = Field(default=0, ge=0)
103         # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
104         court_polygon: Optional[list[list[float]]] = None
105         # Net line for the person two-sided-motion split (person features only ? the shuttle
106         # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
107         net_axis: Literal["x", "y"] = "y"
108         net_position: float = Field(default=0.5, ge=0.0, le=1.0)
109
110
111     class IndexingConfig(BaseModel):
112         default_segmenter: str = "yolo_hybrid"
113         use_gpu: bool = True
114         motion_threshold: float = 0.08

```

```

115     min_segment_duration: float = 3.0
116     dead_time_merge_gap: float = 2.0
117     chunk_duration: float = 90.0
118     chunk_overlap: float = 10.0
119     detector_model: str = "fasterrcnn_resnet50_fpn"
120     detector_score_threshold: float = 0.5
121     yolo_velocity_threshold: float = 0.15
122     yolo_frame_skip: int = 3
123     yolo_tracker: str = "bytetrack.yaml"
124     yolo_prefetch_workers: int = 2
125     min_action_window_duration: float = 2.0
126     log_yolo_candidates: bool = True
127     store_yolo_candidates: bool = True
128     ai_handoff_padding: float = 2.0
129     ai_max_workers: int = 5
130     # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
131     # chunks of a high-bitrate (4K) source blow past the provider upload cap
132     # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
133     # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
134     # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
135     # Matches gemini_refine's --clip-scale-width default.
136     ai_chunk_scale_width: int = 1280
137     # Optional debug knob: trim every video to the first N seconds before processing.
138     debug_duration: Optional[float] = None
139
140     tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
141     wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
142     fusion: FusionConfig = Field(default_factory=FusionConfig)
143
144     @field_validator("default_segmenter")
145     @classmethod
146     def segmenter_must_be_registered(cls, v: str) -> str:
147         # Registry check happens at runtime in main/segmenter selection.
148         # Here we just allow any string for forward compatibility.
149         return v
150
151     @model_validator(mode="after")
152     def _chunk_step_must_be_positive(self) -> "IndexingConfig":
153         # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-
positive
154         # step makes `while t < duration: t += step` loop forever, growing memory before
any
155         # GPU work. Reject the bad config at load time instead.
156         if self.chunk_overlap >= self.chunk_duration:
157             raise ValueError(
158                 f"indexing.chunk_overlap ({self.chunk_overlap}) must be < chunk_duration
"
159                 f"({self.chunk_duration}); otherwise chunking never advances."
160             )
161         return self
162
163
164     class OutputConfig(BaseModel):
165         default_highlights_name: str = "highlights.mp4"
166         db_name: str = "sports_indexer.db"
167         # Directory where the DB, highlight reels, and processed clips are written.
168         # The CLI's --output-dir overrides this at startup (see backend/main.py::main).
169         output_dir: str = "output"
170
171
172     class WatermarkConfig(BaseModel):
173         """Branding overlay burned into each highlight clip during the per-clip re-encode
174         (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
175         (under `stitching.watermark` in config.json) to turn it off. The text is fully
176         configurable. The concat stage stays stream-copy (-c copy)."""

```

```

177     enabled: bool = True
178     text: str = "Generated by Khelsutra.guru"
179     logo_path: str = ""          # optional transparent PNG overlay; "" = no logo
180     font_path: str = ""          # optional .ttf; "" = use the fontconfig `font` family
below
181     font: str = "Sans"           # fontconfig family used when font_path is empty
182     font_size_ratio: float = Field(default=0.035, gt=0.0, le=0.5) # text height as a
fraction of frame height
183     opacity: float = Field(default=0.85, ge=0.0, le=1.0)
184     box: bool = True             # faint backing box behind the text for legibility
185     margin: int = Field(default=24, ge=0) # px inset from the chosen corner
186     position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = "bottom-
right"
187
188
189     class StitchingConfig(BaseModel):
190         begin_padding: float = 0.5
191         end_padding: float = 0.5
192         use_cache: bool = False
193         min_shot_count: int = 1
194         watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
195
196
197     class SecurityConfig(BaseModel):
198         # When set, /api/process video_path must resolve under this directory (hosted/tenant
199         # mode). Default None preserves the single-user local 'any local file' workflow.
200         allowed_video_root: Optional[str] = None
201
202         # --- Chunked upload (B2, PR-2) ---
203         # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
204         # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
205         # contain upload_dir or /api/process will reject the uploaded paths.
206         upload_dir: str = "output/uploads"
207         # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
208         # free-tier edge proxies cap request bodies there (HOSTING_PLAN §2).
209         max_upload_mb: int = 4096
210         max_part_mb: int = 95
211         allowed_upload_extensions: List[str] = ["mp4", "mov"]
212
213
214     class AppConfig(BaseModel):
215         """Root configuration object.
216
217         All runtime configuration should be accessed via an instance of this class
218         (or the loader) rather than raw dict lookups.
219         """
220
221         sport: Sport = "badminton"
222         ai_provider: str = "gemini"
223         # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable
flagship;
224         # the alias never deprecates underneath us. Eval reports record the run date
alongside.
225         ai_model: str = "gemini-flash-latest"
226
227         validation: ValidationConfig = Field(default_factory=ValidationConfig)
228         indexing: IndexingConfig = Field(default_factory=IndexingConfig)
229         output: OutputConfig = Field(default_factory=OutputConfig)
230         stitching: StitchingConfig = Field(default_factory=StitchingConfig)
231         security: SecurityConfig = Field(default_factory=SecurityConfig)
232
233         # Allow extra keys during transition so existing config.json files do not break.
234         model_config = {
235             "extra": "allow",
236             "validate_assignment": True,

```

```

237     }
238
239     # Temporary migration shim for legacy dict-style access (config.get("section",
240     {}).get(...)).
241     # Will be removed once all call sites (validators, segmenters, etc.) move to typed
242     # attribute access / the typed config. A nested *section* is returned as a plain
243     dict
244     # via model_dump, so legacy chained access keeps working ? returning the Pydantic
245     # section model instead breaks it (a model has no .get), which bit every validator
246     on
247     # real runs. Scalar leaves are returned as-is.
248     def get(self, key: str, default: Any = None) -> Any:
249         _MISSING = object()
250         val: Any = _MISSING
251         if key in type(self).model_fields or hasattr(self, key):
252             val = getattr(self, key, _MISSING)
253         if val is _MISSING:
254             # Legacy convenience: find a leaf key inside any nested section. Sections
255             # are
256             # discovered dynamically from the model fields, so a future top-level
257             # section
258             # (e.g. `audio`, `jobs`) is covered automatically without editing a
259             # hardcoded list.
260             for fname in type(self).model_fields:
261                 section = getattr(self, fname, None)
262                 if isinstance(section, BaseModel) and hasattr(section, key):
263                     val = getattr(section, key)
264                     break
265             if val is _MISSING:
266                 return default
267             if isinstance(val, BaseModel):
268                 return val.model_dump(mode="json")
269             return val
270
271     def __getitem__(self, key: str) -> Any:
272         val = self.get(key)
273         if val is None and key not in type(self).model_fields:
274             # Re-raise like a real dict for unknown top-level
275             raise KeyError(key)
276         return val
277
278     # Convenience type alias for callers that want to annotate
279     Config = AppConfig
280
281

```

16. user (2026-06-13T19:50:13.880Z)

```

backend\reporting\harvest.py
backend\reporting\run_signals.py
backend\reporting\spec.yaml
backend\reporting\__init__.py
backend\reporting\__pycache__\harvest.cpython-313.pyc
backend\reporting\__pycache__\run_signals.cpython-313.pyc
backend\reporting\__pycache__\__init__.cpython-313.pyc

```

17. assistant (2026-06-13T19:51:02.584Z)

Let me read the reporting layer, the CSV export tool, and the `\_fmt\_ts` reference to decide the helper's home and signature.

18. user (2026-06-13T19:51:03.535Z)

```

1     """Minimal Value Layer exports (first slice of review item 3).

```

```

2
3 Provides timeline exports (EDL, JSON, CSV) and basic stats over the
4 play_segments returned by the indexer (or query_play_segments). These are
5 the "time-saved editing is table stakes" deliverables and also give a
6 machine-readable foundation for later human correction / golden feeding
7 and clip galleries.
8
9 Kept deliberately small and pure (no DB, no side effects) so they are
10 easy to call from CLI, notebooks, or a future API endpoint.
11 """
12
13 from __future__ import annotations
14
15 from typing import Any, Dict, Iterable, List, Optional
16
17
18 def _to_list(segments: Iterable[Dict[str, Any]]) -> List[Dict[str, Any]]:
19     return list(segments) if not isinstance(segments, list) else segments
20
21
22 def basic_stats(segments: Iterable[Dict[str, Any]]) -> Dict[str, Any]:
23     """Return simple aggregate stats for a list of play segments.
24
25     Keys: count, total_duration_s, avg_duration_s, min_duration_s, max_duration_s.
26     All durations are in seconds (float).
27     """
28     segs = _to_list(segments)
29     if not segs:
30         return {
31             "count": 0,
32             "total_duration_s": 0.0,
33             "avg_duration_s": 0.0,
34             "min_duration_s": 0.0,
35             "max_duration_s": 0.0,
36         }
37     durs = [float(s.get("duration", (s.get("end_time", 0) - s.get("start_time", 0))))
38 for s in segs]
39     total = sum(durs)
40     return {
41         "count": len(segs),
42         "total_duration_s": round(total, 2),
43         "avg_duration_s": round(total / len(durs), 2),
44         "min_duration_s": round(min(durs), 2),
45         "max_duration_s": round(max(durs), 2),
46     }
47
48 def segments_to_json(
49     segments: Iterable[Dict[str, Any]],
50     video_id: str,
51     video_path: Optional[str] = None,
52     meta: Optional[Dict[str, Any]] = None,
53 ) -> Dict[str, Any]:
54     """Return a JSON-serializable timeline dict.
55
56     Shape (stable):
57     {
58         "video_id": "...",
59         "video_path": "...",
60         "meta": {...},
61         "segments": [ {"id":, "start_time":, "end_time":, "duration":, "metadata": {...}},
... ],
62         "stats": basic_stats(...)
63     }
64     """

```



```

65     segs = _to_list(segments)
66     return {
67         "video_id": video_id,
68         "video_path": video_path,
69         "meta": meta or {},
70         "segments": segs,
71         "stats": basic_stats(segs),
72     }
73
74
75 def segments_to_csv(segments: Iterable[Dict[str, Any]], include_header: bool = True) ->
str:
76     """Return a CSV string (start,end,duration,shot_count,detector,...) .
77
78     Compatible with many editors and the existing golden/slicer schemas
79     (extra columns are harmless).
80     """
81     segs = _to_list(segments)
82     lines: List[str] = []
83     if include_header:
84
85 lines.append("start_time,end_time,duration,shot_count,shot_count_confidence,detector,id")
86     for s in segs:
87         md = s.get("metadata") or {}
88         shot = md.get("shot_count", "")
89         conf = md.get("shot_count_confidence", "")
90         det = md.get("detector", "")
91         lines.append(
92             f"{s.get('start_time',0)},{s.get('end_time',0)},{s.get('duration',0)},{s.get('shot_count',0)},{s.get('shot_count_confidence',0)},{s.get('detector','')},{s.get('id','')}"
93         )
94     return "\n".join(lines) + "\n"
95
96
97 def segments_to_edl(segments: Iterable[Dict[str, Any]], title: str = "Highlights") ->
str:
98     """Return a simple EDL (CMX 3600-ish) for the segments.
99
100     Many NLEs will ingest this for a rough cut. Uses record timecode 00:00:00:00
101     as origin; clips are placed sequentially with the original source in/out.
102     """
103     segs = _to_list(segments)
104     lines: List[str] = [f"TITLE: {title}", "FCM: NON-DROP FRAME"]
105     for i, s in enumerate(segs, 1):
106         start = float(s.get("start_time", 0.0))
107         end = float(s.get("end_time", start))
108         # EDL is HH:MM:SS:FF (assume 30fps for the timecode display; real edit rate
109         # is irrelevant for a rough cut ? the numbers are just labels here).
110         def _tc(t: float) -> str:
111             h = int(t // 3600)
112             m = int((t % 3600) // 60)
113             s_ = int(t % 60)
114             f = int((t - int(t)) * 30) # 30 fps display
115             return f"{h:02d}:{m:02d}:{s_:02d}:{f:02d}"
116         rec_in = _tc((i - 1) * 0.0) # place back-to-back in the EDL timeline
117         rec_out = _tc((i - 1) * 0.0 + (end - start))
118         src_in = _tc(start)
119         src_out = _tc(end)
120         lines.append(f"{i:03d}  AX  V      C          {rec_in} {rec_out} {src_in} {src_out}")
121         lines.append(f"* FROM CLIP NAME: {title} {i}")
122         lines.append("")
123     return "\n".join(lines)
124

```

19. user (2026-06-13T19:51:04.477Z)

```
1      """In-process signal channel: pipeline internals -> the per-video report.
2
3      Some report-worthy facts (e.g. the provider refusing an oversize chunk) surface
4      deep inside a segmenter's worker threads, with no return path to the harvest ?
5      segmenters return only their segment list. Layers record those facts here keyed
6      by video_id; `emit_indexer_report` pops them when the run's report is built.
7
8      Process-local and best-effort BY DESIGN: recording must never break a run, the
9      store is consumed (popped) once per report, and signals for a video are simply
10     dropped if no report is emitted. No obsreport dependency ? importable everywhere.
11     """
12     from __future__ import annotations
13
14     import threading
15     from typing import Any, Dict
16
17     _lock = threading.Lock()
18     _signals: Dict[str, Dict[str, Any]] = {}
19
20
21     def incr(video_id: str, key: str, by: int = 1) -> None:
22         """Add `by` to the counter `key` for this video's in-flight run."""
23         if not video_id or by == 0:
24             return
25         with _lock:
26             sig = _signals.setdefault(video_id, {})
27             sig[key] = int(sig.get(key, 0)) + by
28
29
30     def pop(video_id: str) -> Dict[str, Any]:
31         """Consume (remove and return) all signals recorded for this video's run."""
32         with _lock:
33             return _signals.pop(video_id, {})
34
```

20. user (2026-06-13T19:51:05.138Z)

```
30     from backend.sports import sport_registry
31     from backend.providers import provider_registry
32     from backend.pipeline.detectors.base import DetectorRunner
33
34     logger = logging.getLogger(__name__)
35
36
37     def _fmt_ts(seconds: float) -> str:
38         seconds = max(0, int(seconds))
39         h, rem = divmod(seconds, 3600)
40         m, s = divmod(rem, 60)
41         return f"{h:02d}:{m:02d}:{s:02d}"
42
43
44     class TrajectoryHybridSegmenter(BasePlaySegmenter):
45         """Trajectory-detector hybrid: precise candidate rallies + AI semantic
46         boundaries."""
47
48         #: config section under `indexing` (velocity_threshold lives there), the
49         #: output subdir for trajectories, and the metadata detector-tag prefix.
50         family: str = ""
```

21. user (2026-06-13T19:51:57.515Z)

```
backend\utils\__init__.py
backend\utils\geometry.py
```

```
backend\utils\hashing.py
backend\utils\paths.py
backend\utils\video.py
```

22. assistant (2026-06-13T19:51:55.985Z)

Now let me look at the test structure and confirm the `utils` package + how tests are discovered.

23. user (2026-06-13T19:51:56.938Z)

```
1      #!/usr/bin/env python
2      """Run the FULL test suite for the badminton-highlight-indexer.
3
4      A single, discoverable entry point so you (or a reviewer / CI) can verify the
5      whole suite before opening a PR:
6
7          python run_all_tests.py          # full suite, quiet
8          python run_all_tests.py -v       # verbose (passthrough to pytest)
9          python run_all_tests.py -k config # filter by keyword (passthrough)
10         python run_all_tests.py --cov=backend --cov-report=term-missing # coverage
11
12     Anything after the script name is passed straight through to pytest.
13
14     Notes
15     -----
16     - The suite is fully mocked (no GPU, no network, no WSL) and runs offline.
17     - The observability **reporting** tests skip cleanly if the optional ``obsreport``
18       package isn't installed (they use ``pytest.importorskip``). To exercise them too:
19         pip install -e ../sports-obsreport --no-build-isolation
20     - Exit code is pytest's own, so this is CI / pre-PR friendly (0 = all green).
21     """
22     from __future__ import annotations
23
24     import importlib.util
25     import subprocess
26     import sys
27
28
29     def _installed(module: str) -> bool:
30         try:
31             return importlib.util.find_spec(module) is not None
32         except (ImportError, ValueError):
33             return False
34
35
36     def main() -> int:
37         args = list(sys.argv[1:])
38         # Default to quiet unless the caller chose a verbosity.
39         if not any(a in ("-q", "-v", "-vv", "-vvv", "--quiet", "--verbose") for a in args):
40             args = ["-q", *args]
41         cmd = [sys.executable, "-m", "pytest", *args]
42
43         obs = "available" if _installed("obsreport") else "NOT installed ? reporting tests
will SKIP"
44         bar = "=" * 72
45         print(bar)
46         print("Full test suite: " + " ".join(cmd))
47         print(f"obsreport (optional reporting dep): {obs}")
48         print(bar, flush=True)
49
50         rc = subprocess.run(cmd).returncode
51
52         print(bar)
53         print("RESULT: " + ("PASS - all tests green" if rc == 0 else f"FAIL - pytest exit
```

```
{rc}"))
54         print(bar)
55         return rc
56
57
58     if __name__ == "__main__":
59         raise SystemExit(main())
60
```

24. user (2026-06-13T19:51:57.587Z)

tests\test_export_wasb_segments.py

25. assistant (2026-06-13T19:52:00.301Z)

Let me check existing tests for the export tool and config to match test style.

26. user (2026-06-13T19:52:00.873Z)

tests__init__.py
tests\test_eval_batch.py
tests\test_eval_training.py
tests\test_annotation_providers.py
tests\test_calibration.py
tests\test_calibrate_wasb.py
tests\test_export_wasb_segments.py
tests\test_gemini_refine.py
tests\test_calibrate_local.py
tests\test_distill_local.py
tests\test_automated_provider.py
tests\test_eval_harness.py
tests\test_hit_detector.py
tests\test_validators_base.py
tests\test_hashing.py
tests\test_rally_slicer.py
tests\test_reporting_spec_faq.py
tests\test_tracknet_runner.py
tests\test_eval_partial_labels.py
tests\test_detector_keying.py
tests\test_reingest_safety.py
tests\test_eval_metrics.py
tests\test_compiler.py
tests\test_eval_annotations.py
tests\test_gt_loader.py
tests\test_import_smoke.py
tests\test_config_wasb.py
tests\test_reporting_harvest.py
tests\test_run_signals.py
tests\test_async_jobs.py
tests\test_base.py
tests\test_database.py
tests\test_eval_classifier.py
tests\test_eval_regression.py
tests\test_gemini.py
tests\test_geometry.py
tests\test_path_safety.py
tests\test_provider_gemini.py
tests\test_providers.py
tests\test_rally_seq_proto.py
tests\test_reliability_hardening.py
tests\test_reporting.py
tests\test_sport_profiles.py
tests\test_training_gen0.py
tests\test_video.py

```
tests\test_wasb_cache.py
tests\test_tracknet_hybrid.py
tests\test_wsl_tilde_quoting.py
tests\test_upload_download.py
tests\test_api_endpoints.py
tests\test_watermark.py
tests\test_cleanup_caches.py
tests\test_wasb_runner.py
tests\test_experiment.py
tests\test_yolo_hybrid.py
tests\test_config.py
tests\test_fusion_features.py
tests\test_fusion_hybrid.py
tests\test_person_detector.py
tests\test_rally_gate.py
tests\test_trajectory_hybrid_equivalence.py
```

27. user (2026-06-13T19:52:01.663Z)

Found 2 files

```
tests\test_rally_slicer.py
tests\test_export_wasb_segments.py
```

28. user (2026-06-13T19:52:08.269Z)

```
1      """Basic tests for the typed configuration package.
2
3      These ensure:
4      - Defaults are sensible and come from the models.
5      - File overrides work.
6      - Old config files with extra keys are tolerated.
7      - save_default_config produces loadable output.
8      """
9
10     import json
11     import tempfile
12     import os
13
14
15     from backend.config import load_config, save_default_config, AppConfig
16
17
18     def test_load_default_config():
19         # Force load without the project's config.json so we test the model built-in
20         defaults
21         cfg = load_config("/nonexistent/path/that/does/not/exist/12345/config.json")
22         assert isinstance(cfg, AppConfig)
23         assert cfg.sport == "badminton"
24         assert cfg.indexing.default_segmenter == "yolo_hybrid"
25         assert cfg.validation.min_blur_threshold == 20.0 # updated default from review
26         assert cfg.indexing.ai_max_workers == 5
27
28     def test_fusion_config_defaults_reproduce_control():
29         """The fusion block defaults must be no-ops: wasb substrate, no cues, no gating, no
30         mask."""
31         cfg = load_config("/nonexistent/path/12345/config.json")
32         f = cfg.indexing.fusion
33         assert f.substrate == "wasb"
34         assert f.cue_flags == {} # person cue OFF
35         assert f.min_crossings == 0 # Gate A OFF
36         assert f.min_players == 0 # Gate B OFF
37         assert f.court_polygon is None # no mask
38         assert f.velocity_threshold == 0.02
```

```

39
40 def test_fusion_config_file_override():
41     with tempfile.TemporaryDirectory() as td:
42         p = os.path.join(td, "fusion.json")
43         with open(p, "w") as fh:
44             json.dump({"indexing": {"fusion": {
45                 "substrate": "tracknet", "min_crossings": 2,
46                 "cue_flags": {"person": True}, "court_polygon": [[0.1, 0.1], [0.9, 0.1],
[0.5, 0.9]],
47                 }}, fh)
48         cfg = load_config(p)
49         f = cfg.indexing.fusion
50         assert f.substrate == "tracknet"
51         assert f.min_crossings == 2
52         assert f.cue_flags == {"person": True}
53         assert len(f.court_polygon) == 3
54         # An old config without a fusion block still loads (defaults applied).
55         assert cfg.indexing.wasb.velocity_threshold == 0.02
56
57
58 def test_file_override():
59     with tempfile.TemporaryDirectory() as td:
60         p = os.path.join(td, "override.json")

```

29. user (2026-06-13T19:52:15.363Z)

```

1     """Configuration loader and saver.
2
3     Provides a single place to load validated AppConfig with built-in defaults
4     merged with optional file overrides (config.json).
5
6     This replaces scattered json.load + .get() patterns throughout the codebase.
7     """
8
9     from __future__ import annotations
10
11     import json
12     import logging
13     from pathlib import Path
14     from typing import Any
15
16     from pydantic import BaseModel
17
18     from .models import AppConfig
19
20     logger = logging.getLogger(__name__)
21
22     DEFAULT_CONFIG_PATH = Path("config.json")
23
24
25     def _unknown_key_paths(data: dict[str, Any], model_cls: type[BaseModel], prefix: str =
"" -> list[str]:
26         """Dotted paths in `data` that the typed config will not honor.
27
28         Two silent failure modes exist without this check: a typo'd TOP-LEVEL key is
29         kept by AppConfig's ``extra="allow"`` but never read by typed code, and a
30         typo'd key INSIDE a section is silently dropped by Pydantic. Either way the
31         user's intent is lost ? collect every such key so load_config can warn.
32         """
33         unknown: list[str] = []
34         fields = model_cls.model_fields
35         for key, val in data.items():
36             if key not in fields:
37                 unknown.append(prefix + key)
38                 continue

```

```

39         ann = fields[key].annotation
40         if isinstance(val, dict) and isinstance(ann, type) and issubclass(ann,
BaseModel):
41             unknown.extend(_unknown_key_paths(val, ann, f"{prefix}{key}.")
42         return unknown
43
44
45 def _get_builtin_defaults() -> dict[str, Any]:
46     """Return a plain dict of the current built-in defaults.
47
48     The AppConfig model is the single source of truth for defaults.
49     """
50     return AppConfig().model_dump(mode="json")
51
52
53 def load_config(path: str | Path | None = None) -> AppConfig:
54     """Load configuration with the following precedence:
55
56     1. Built-in defaults defined in the Pydantic models.
57     2. Values from the JSON file (if present and readable).
58     3. (Future) environment / CLI overrides.
59
60     Missing file or unreadable file ? returns a fully defaulted AppConfig.
61     Unknown keys in the file are tolerated (extra="allow" on the model).
62     """
63     cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
64
65     file_data: dict[str, Any] = {}
66     if cfg_path.exists():
67         try:
68             with open(cfg_path, "r", encoding="utf-8") as f:
69                 file_data = json.load(f)
70         except Exception as exc:
71             # Non-fatal: continue with defaults + whatever we could parse.
72             # In production we might log here.
73             logger.warning(f"Failed to read {cfg_path}: {exc}. Using defaults + partial
data.")
74
75     if file_data:
76         unknown = _unknown_key_paths(file_data, AppConfig)
77         if unknown:
78             logger.warning(
79                 "[CONFIG WARNING] %s contains keys the typed config does not "
80                 "recognize (typo'd or removed knobs ? top-level extras are kept "
81                 "but unread; unknown section keys are silently dropped): %s",
82                 cfg_path, ", ".join(sorted(unknown)),
83             )
84
85     # Merge: start from defaults, overlay file data (Pydantic will validate/coerce).
86     merged = {**_get_builtin_defaults(), **file_data}
87
88     return AppConfig.model_validate(merged)
89
90
91 def save_default_config(path: str | Path | None = None) -> Path:
92     """Write a fresh default config.json.
93
94     Used by setup.py diagnostics and the --setup CLI path.
95     Overwrites if the file exists.
96     """
97     cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
98     cfg = AppConfig()
99     cfg_path.parent.mkdir(parents=True, exist_ok=True)
100     cfg_path.write_text(cfg.model_dump_json(indent=4), encoding="utf-8")
101     return cfg_path

```

```

102
103
104     # Backwards-compatible alias used by some older call sites during transition.
105     def get_default_config() -> dict[str, Any]:
106         """Return defaults as a plain dict (for transitional use only)."""
107         return _get_builtin_defaults()
108

```

30. user (2026-06-13T19:53:15.264Z)

```

1     # Utils package
2

```

31. assistant (2026-06-13T19:53:14.044Z)

I have a complete picture. Let me verify `backend/\_\_init\_\_.py` is lightweight (so importing the logging helper early is safe) and confirm there's no existing `test\_export.py`.

32. user (2026-06-13T19:53:14.997Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer.

33. assistant (2026-06-13T19:53:20.635Z)

Branch is safe to set up. Let me create it from the **remote** master (not the current working branch).

34. user (2026-06-13T19:53:21.580Z)

```

branch 'feat/runlog-readability' set up to track 'origin/master'.
Switched to a new branch 'feat/runlog-readability'
feat/runlog-readability
35a50f3 docs(readme): index QUALITY_ITERATIONS + mark EXPERIMENT_HARNESS P1 SHIPPED (#118)

```

35. user (2026-06-13T19:53:47.259Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\logging_setup.py (file state is current in your context ? no need to Read it back)

36. assistant (2026-06-13T19:53:36.663Z)

Branch is correctly based on `origin/master` @ 35a50f3. Here's my plan before I implement:

Plan

```

**Change 1 ? quieter `run.log` (HTTP chatter ? WARNING, config-gated)**
- New central helper `backend/utils/logging_setup.py::setup_logging(verbose_http=False)` ? runs
`basicConfig` (same format as today) and raises `httpx, httpcore, urllib3, google_genai,
google.genai, google.auth, grpc` to `WARNING` (NOTSET when `verbose_http=True` to restore for
debugging). Idempotent; never touches `backend.*`.
- New config knob `logging.verbose_http: bool = False` (`LoggingConfig` on `AppConfig`).
- Wire into the manual-QA run path `run_process_and_stitch.py`: call `setup_logging()` at import
(quiet default), then `setup_logging(verbose_http=cfg.logging.verbose_http)` right after config
load.

**Change 2 ? end-of-run rally-times summary**
- New pure helper `backend/tools/export.py::format_rally_summary(segments) -> str` (sits next to
`segments_to_csv`/`basic_stats`, same play_segment dicts ? text): header (total rallies + total
play in HH:MM:SS) + per-rally table `# start end dur(s) shot_count`, HH:MM:SS via a local
`_fmt_hms` matching the segmenters' `_fmt_ts`. Handles `metadata` as dict or JSON string.
- Log it as one block at the END of `run_process_and_stitch.py::main()` so the log lines up row-

```


for-row with the rally CSV.

****Tests:**** `tests/test\_logging\_setup.py`, `tests/test\_rally\_summary.py`, + a default/override assertion in `tests/test\_config.py`. Verify `python run\_all\_tests.py` green, ruff + mypy clean.

Implementing now.

37. assistant (2026-06-13T19:53:47.234Z)

```
"""Central logging setup for the indexer's CLI/run entrypoints.
```

```
One place to (a) install the root log format every entrypoint shares and (b) tame
the third-party HTTP/RPC client libraries that log one INFO line per request. On the
Gemini path (`google-genai` -> `httpx`/`grpc`/`google.auth`) that is dozens of
lines per run, which buries our own `[rally]`/`[compile]`/`[CACHE]` output in
`run.log` ? the very thing the owner eyeballs during manual QA.
```

```
The suppression is config-gated (`logging.verbose_http`) so request-flow debugging can
restore the chatter without code edits, and it only ever touches the named third-party
loggers ? our own `backend.*` loggers are left at the root level untouched.
"""
```

```
from __future__ import annotations
```

```
import logging
```

```
#: Third-party client loggers that emit per-request INFO chatter. Raised to WARNING by
#: default so only genuine warnings/errors from them reach run.log.
```

```
NOISY_HTTP_LOGGERS = (
    "httpx",
    "httpcore",
    "urllib3",
    "google_genai", # google-genai SDK (underscore form)
    "google.genai", # dotted form, depending on SDK version
    "google.auth",
    "grpc",
)
```

```
_LOG_FORMAT = "%(asctime)s [%(name)s] %(levelname)s: %(message)s"
```

```
_DATE_FORMAT = "%H:%M:%S"
```

```
def setup_logging(verbose_http: bool = False, level: int = logging.INFO) -> None:
```

```
    """Configure root logging and (un)suppress noisy third-party HTTP loggers.
```

```
Idempotent and safe to call more than once: `basicConfig` installs the root
handler/format only the first time, while the third-party levels are (re)applied on
every call ? so an entrypoint can call this early with the quiet default and again
after config load to honour `logging.verbose_http`.
```

```
Args:
```

```
    verbose_http: when True, restore full per-request chatter (levels set to
        NOTSET so they inherit the root level); when False (default), raise the
        noisy loggers to WARNING.
```

```
    level: root logger level for our own `backend.*` logs (default INFO).
```

```
"""
```

```
logging.basicConfig(level=level, format=_LOG_FORMAT, datefmt=_DATE_FORMAT)
```

```
http_level = logging.NOTSET if verbose_http else logging.WARNING
```

```
for name in NOISY_HTTP_LOGGERS:
```

```
    logging.getLogger(name).setLevel(http_level)
```

38. assistant (2026-06-13T19:53:49.413Z)

Now the config knob:

39. user (2026-06-13T19:53:52.206Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py has been updated successfully. (file state is current in your context ? no need to Read it back)

40. user (2026-06-13T19:53:56.267Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py has been updated successfully. (file state is current in your context ? no need to Read it back)

41. assistant (2026-06-13T19:53:59.977Z)

Now the rally-summary helper in `export.py`:

42. user (2026-06-13T19:54:10.893Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py has been updated successfully. (file state is current in your context ? no need to Read it back)

43. assistant (2026-06-13T19:54:14.013Z)

Now wire both into the run path. First the logging setup at import + after config load:

44. user (2026-06-13T19:54:18.913Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py has been updated successfully. (file state is current in your context ? no need to Read it back)

45. user (2026-06-13T19:54:25.505Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py has been updated successfully. (file state is current in your context ? no need to Read it back)

46. assistant (2026-06-13T19:54:28.015Z)

Now add the summary block at the end of the run. Let me view the current end of `main()`:

47. user (2026-06-13T19:54:28.849Z)

```
128         if skipped_count > 0:
129             print(f"[FILTER] Excluded {skipped_count} play segments with fewer than
{min_shot_count} shots.", flush=True)
130         print(f"[FILTER] Retaining {len(time_pairs)} play segments for stitching.",
flush=True)
131
132         if not time_pairs:
133             print("No play segments passed the filtering criteria. Cannot compile
highlights.")
134             return
135
136         compiler = VideoCompiler(output_dir, end_padding=end_padding,
begin_padding=begin_padding,
137                                 watermark=cfg.stitching.watermark)
138         try:
139             out_path = compiler.compile_highlights(video_path, time_pairs, output_filename)
140             print(f"Stitched highlight video saved to: {out_path}", flush=True)
141         except Exception as e:
142             print(f"Error during stitching: {e}", flush=True)
143
144     if __name__ == "__main__":
145         main()
146
```

48. assistant (2026-06-13T19:54:56.675Z)

I'll emit the summary as the final block (after stitching, so ffmpeg output doesn't bury it), and also on the all-filtered early-return path so a fully-filtered run still shows what was produced:

49. user (2026-06-13T19:55:04.151Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py has been updated successfully. (file state is current in your context ? no need to Read it back)

50. assistant (2026-06-13T19:55:06.553Z)

Now the tests. First the rally-summary helper test:

51. assistant (2026-06-13T19:55:15.738Z)

"""Tests for backend.tools.export.format_rally_summary ? the end-of-run rally table the owner eyeballs against the rally CSV during manual QA.

Pure/offline: feeds play_segment dicts in, asserts on the rendered text.
"""

```
from backend.tools.export import format_rally_summary
```

```
def _seg(start, end, shot_count=None, metadata=None):
    seg = {"start_time": start, "end_time": end, "duration": end - start}
    if metadata is not None:
        seg["metadata"] = metadata
    elif shot_count is not None:
        seg["metadata"] = {"shot_count": shot_count}
    return seg
```

```
def test_header_counts_and_total_duration():
    segs = [_seg(10, 25, 8), _seg(40, 58, 11)] # 15s + 18s = 33s total
    out = format_rally_summary(segs)
    header = out.splitlines()[0]
    assert "2 rallies" in header
    assert "00:00:33" in header # total play in HH:MM:SS
    assert "(33.0s)" in header # and in seconds
```

```
def test_per_rally_rows_use_hms_and_shot_count():
    segs = [_seg(10, 25, 8), _seg(3661, 3680, 11)] # second crosses the hour mark
    rows = format_rally_summary(segs).splitlines()
    # header + column header + 2 data rows
    assert len(rows) == 4
    assert rows[2].split() == ["1", "00:00:10", "00:00:25", "15.0", "8"]
    assert rows[3].split() == ["2", "01:01:01", "01:01:20", "19.0", "11"]
```

```
def test_missing_shot_count_renders_dash():
    out = format_rally_summary([_seg(0, 5)]) # no metadata at all
    assert out.splitlines()[2].split() == ["1", "00:00:00", "00:00:05", "5.0", "-"]
```

```
def test_metadata_as_json_string_is_parsed():
    # DB round-trip stores metadata as a JSON string; the summary must still read shot_count.
    seg = _seg(0, 5, metadata='{"shot_count": 7}')
    assert out_last_field(format_rally_summary([seg])) == "7"
```

```
def test_empty_segments():
    out = format_rally_summary([])
    assert "0 rallies" in out.splitlines()[0]
    assert "00:00:00" in out.splitlines()[0]
    assert "(no rallies detected)" in out
```

```
def out_last_field(summary: str) -> str:
    return summary.splitlines()[2].split()[-1]
```

52. user (2026-06-13T19:55:15.752Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_rally_summary.py (file state is current in your context ? no need to Read it back)

53. assistant (2026-06-13T19:55:24.399Z)

```
"""Tests for backend.utils.logging_setup.setup_logging ? quieting third-party HTTP
chatter in run.log behind the logging.verbose_http knob, without touching our own
backend.* loggers.
"""
```

```
import logging
```

```
import pytest
```

```
from backend.utils.logging_setup import NOISY_HTTP_LOGGERS, setup_logging
```

```
@pytest.fixture
```

```
def restore_noisy_levels():
    """Save/restore the noisy loggers' levels so these tests don't leak global state."""
    saved = {name: logging.getLogger(name).level for name in NOISY_HTTP_LOGGERS}
    try:
        yield
    finally:
        for name, lvl in saved.items():
            logging.getLogger(name).setLevel(lvl)
```

```
def test_default_raises_noisy_http_loggers_to_warning(restore_noisy_levels):
    setup_logging() # quiet default
    for name in NOISY_HTTP_LOGGERS:
        assert logging.getLogger(name).level == logging.WARNING, name
```

```
def test_verbose_http_restores_chatter(restore_noisy_levels):
    setup_logging() # suppress first
    setup_logging(verbose_http=True) # then restore
    for name in NOISY_HTTP_LOGGERS:
        # NOTSET => inherit the root level (INFO), i.e. chatter shows again.
        assert logging.getLogger(name).level == logging.NOTSET, name
```

```
def test_does_not_touch_backend_loggers(restore_noisy_levels):
    # Our own loggers must keep emitting at INFO ? we only ever tune the named third parties.
    setup_logging()
    assert logging.getLogger("backend.pipeline.segmenters.trajectory_hybrid").level ==
logging.NOTSET
    assert logging.getLogger().isEnabledFor(logging.INFO)
```

54. user (2026-06-13T19:55:24.410Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_logging_setup.py (file state is current in your context ? no need to Read it back)

55. assistant (2026-06-13T19:55:27.177Z)

Now add config assertions for the new knob to `test\_config.py`:

56. user (2026-06-13T19:55:30.661Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py has been updated successfully. (file state is current in your context ? no need to Read it back)

57. assistant (2026-06-13T19:55:34.699Z)

Now let me run the new/affected tests and the linters.

58. user (2026-06-13T19:55:36.828Z)

```
.....F..... [100%]
===== FAILURES =====
_____ test_does_not_touch_backend_loggers _____

restore_noisy_levels = None

    def test_does_not_touch_backend_loggers(restore_noisy_levels):
        # Our own loggers must keep emitting at INFO ? we only ever tune the named third
parties.
        setup_logging()
        assert logging.getLogger("backend.pipeline.segmenters.trajectory_hybrid").level ==
logging.NOTSET
>     assert logging.getLogger().isEnabledFor(logging.INFO)
E       assert False
E         + where False = isEnabledFor(20)
E         +   where isEnabledFor = <RootLogger root (WARNING)>.isEnabledFor
E         +     where <RootLogger root (WARNING)> = <function getLogger at 0x00000272EDEC1A80>()
E         +       where <function getLogger at 0x00000272EDEC1A80> = logging.getLogger
E         +         and    20 = logging.INFO

tests\test_logging_setup.py:42: AssertionError
===== short test summary info =====
FAILED tests/test_logging_setup.py::test_does_not_touch_backend_loggers - ass...
1 failed, 20 passed in 0.61s
```

59. assistant (2026-06-13T19:55:56.358Z)

The root-level assertion fails only under pytest (pytest pre-installs a root handler, so `basicConfig` becomes a no-op for level). In the real entrypoint `basicConfig` runs first and does set INFO ? but the `level` param should take effect deterministically regardless. I'll force the root level explicitly (more robust) and make the test restore it.

60. user (2026-06-13T19:55:59.526Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\logging_setup.py has been updated successfully. (file state is current in your context ? no need to Read it back)

61. user (2026-06-13T19:56:04.927Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_logging_setup.py has been updated successfully. (file state is current in your context ? no need to Read it back)

62. user (2026-06-13T19:56:09.345Z)

.....
21 passed in 0.35s

[100%]

63. assistant (2026-06-13T19:56:16.785Z)

All green. Now ruff + mypy on the touched files, then the full suite.

64. user (2026-06-13T19:56:18.087Z)

All checks passed!

65. user (2026-06-13T19:56:22.997Z)

Success: no issues found in 3 source files

66. assistant (2026-06-13T19:56:28.090Z)

Let me check how CI runs mypy/ruff (to match scope) and run the full suite.

67. user (2026-06-13T19:56:30.308Z)

```
.github/workflows/ci.yml
mypy.ini
ruff.toml
---- CI mypy/ruff invocation ----
.github/workflows/ci.yml: # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle + bugbear.
.github/workflows/ci.yml: # Also the repo-wide syntax guard ? ruff reports invalid-syntax on
every
.github/workflows/ci.yml:     name: Ruff lint
.github/workflows/ci.yml:     - name: Install ruff
.github/workflows/ci.yml:       run: python -m pip install ruff
.github/workflows/ci.yml:     - name: Ruff check
.github/workflows/ci.yml:       run: ruff check . --output-format github
.github/workflows/ci.yml: # Type gate (mypy.ini at repo root): lenient green baseline with an
explicit
.github/workflows/ci.yml: # up. Typed core deps are installed so mypy sees their real
signatures
.github/workflows/ci.yml:     name: Mypy (lenient baseline)
.github/workflows/ci.yml:     - name: Install mypy + typed core deps
.github/workflows/ci.yml:       run: python -m pip install mypy pydantic fastapi uvicorn
python-dotenv google-genai numpy
.github/workflows/ci.yml:     - name: Mypy
.github/workflows/ci.yml:       run: mypy backend training
```

68. assistant (2026-06-13T19:56:35.769Z)

CI runs `ruff check .` (whole repo) and `mypy backend training`. Let me match those exactly, plus the full test suite.

69. user (2026-06-13T19:56:41.091Z)

All checks passed!

==== MYPY ====

```
backend\annotations\base.py:85: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\sports\__init__.py:7: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs [annotation-unchecked]
backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\providers\__init__.py:7: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 81 source files
```

70. assistant (2026-06-13T19:56:43.860Z)

Ruff clean, mypy clean (the notes are pre-existing baseline noise, "no issues found"). Now the full suite:

71. user (2026-06-13T19:57:23.648Z)

```
=====
Full test suite: C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
-q
obsreport (optional reporting dep): available
=====
..... [ 15%]
..... [ 30%]
..... [ 45%]
..... [ 60%]
..... [ 75%]
..... [ 90%]
..... [100%]
477 passed in 36.35s
=====
RESULT: PASS - all tests green
=====
```

72. assistant (2026-06-13T19:57:29.656Z)

All 477 pass. Let me do a quick functional sanity check of the summary output and confirm the entrypoint compiles:

73. user (2026-06-13T19:57:32.325Z)

```
=== Rally summary: 3 rallies, 01:00:05 (3605.4s) total play ===
#  start      end      dur(s)  shots
1  00:00:10  00:00:25    15.4     8
2  00:01:35  01:01:20  3585.0    11
3  01:01:40  01:01:45     5.0     -
---- compile entrypoint ----
run_process_and_stitch.py compiles OK
```

74. assistant (2026-06-13T19:57:41.499Z)

Output is clean and aligned, JSON-string metadata parsed (rally 2 ? 11 shots), missing shot_count ? `.`. Now let me re-verify branch state and stage only my explicit paths.

75. user (2026-06-13T19:57:43.074Z)

```
feat/runlog-readability
35a50f3 docs(readme): index QUALITY_ITERATIONS + mark EXPERIMENT_HARNESS P1 SHIPPED (#118)
---- my changed files ----
M backend/config/models.py
M backend/tools/export.py
M run_process_and_stitch.py
M tests/test_config.py
?? backend/utils/logging_setup.py
?? tests/test_logging_setup.py
?? tests/test_rally_summary.py
```

76. user (2026-06-13T19:57:48.790Z)

warning: in the working copy of 'backend/utils/logging_setup.py', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'tests/test_logging_setup.py', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'tests/test_rally_summary.py', LF will be replaced by CRLF the

```

next time Git touches it
---- staged ----
M backend/config/models.py
M backend/tools/export.py
A backend/utis/logging_setup.py
M run_process_and_stitch.py
M tests/test_config.py
A tests/test_logging_setup.py
A tests/test_rally_summary.py
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
?? scratch/arial.ttf
?? scratch/debug_wsl.py
?? scratch/keep_awake.ps1
?? scratch/probe_cv2_count.py
?? scratch/probe_gemini.py
?? scratch/repro_drawtext_font.py
?? scratch/repro_movie_overlay.py
?? scratch/run_gx010125.py
?? scratch/stitch_min5.py
?? scratch/trim_video.py
---- diffstat ----
 backend/config/models.py      | 10 ++++++
 backend/tools/export.py       | 59 ++++++++++++++++++++++++++++++++++++++
 backend/utis/logging_setup.py | 56 ++++++++++++++++++++++++++++++++++++++
 run_process_and_stitch.py     | 25 ++++++-----
 tests/test_config.py          | 15 ++++++
 tests/test_logging_setup.py   | 44 ++++++++++++++++++++++++++++++++++
 tests/test_rally_summary.py   | 56 ++++++++++++++++++++++++++++++++++++++
 7 files changed, 259 insertions(+), 6 deletions(-)

```

77. assistant (2026-06-13T19:57:55.843Z)

Only my 7 files are staged; the sibling session's `docs/`, `artifacts/`, and `scratch/` changes are untouched. Committing now.

78. user (2026-06-13T19:58:03.272Z)

```

[feat/runlog-readability 80643b8] @ feat(logging): quieter run.log ? suppress HTTP chatter +
end-of-run rally summary
 7 files changed, 259 insertions(+), 6 deletions(-)
 create mode 100644 backend/utis/logging_setup.py
 create mode 100644 tests/test_logging_setup.py
 create mode 100644 tests/test_rally_summary.py

```

79. assistant (2026-06-13T19:58:13.560Z)

The commit message got mangled ? I used PowerShell here-string syntax (`@'...'@`) inside the ****Bash**** tool, which treats `@` literally. Let me inspect and fix it.

80. user (2026-06-13T19:58:14.558Z)

```

@$
feat(logging): quieter run.log M-bM-^@M-^T suppress HTTP chatter + end-of-run rally summary$
$
Two manual-QA readability changes for run.log (the owner runs the tool on a$
video, then eyeballs the rally CSV against the log to verify quality).$

```

81. assistant (2026-06-13T19:58:22.881Z)

Confirmed ? leading `@` line and trailing `@`. I'll fix it cleanly by writing the message to a temp file and amending with `-F`.

82. assistant (2026-06-13T19:58:29.749Z)

feat(logging): quieter run.log ? suppress HTTP chatter + end-of-run rally summary

Two manual-QA readability changes for run.log (the owner runs the tool on a video, then eyeballs the rally CSV against the log to verify quality).

- 1) Suppress third-party HTTP/RPC chatter. The Gemini path (google-genai -> httpx/grpc/google.auth) logs one INFO line per request, burying our own [rally]/[compile]/[CACHE] lines. New central helper backend/utils/logging_setup.py::setup_logging raises httpx, httpcore, urllib3, google_genai/google.genai, google.auth, grpc to WARNING by default, gated behind a new config knob logging.verbose_http (false). Our own backend.* loggers are untouched. Wired into run_process_and_stitch.py.
- 2) End-of-run rally-times summary. New pure helper backend/tools/export.py::format_rally_summary(segments) -> str renders a header (total rallies + total play in HH:MM:SS) plus a per-rally table (# start end dur(s) shot_count) from the produced play_segments, so the log lines up row-for-row with the rally CSV. Logged as one block at the end of run_process_and_stitch.py::main(). Timestamps use HH:MM:SS to match _fmt_ts.

Tests: tests/test_logging_setup.py, tests/test_rally_summary.py, + logging.verbose_http default/override in tests/test_config.py.
Full suite green (477), ruff + mypy clean. Default behavior = quieter log + the summary block.

Co-Authored-By: Claude Opus 4.8 (1M context) <noreply@anthropic.com>

83. user (2026-06-13T19:58:29.766Z)

File created successfully at: C:\Users\avidu\AppData\Local\Temp\runlog_commit_msg.txt (file state is current in your context ? no need to Read it back)

84. user (2026-06-13T19:58:34.525Z)

```
[feat/runlog-readability 3af7d1a] feat(logging): quieter run.log ? suppress HTTP chatter + end-of-run rally summary
Date: Sun Jun 14 01:28:03 2026 +0530
7 files changed, 259 insertions(+), 6 deletions(-)
create mode 100644 backend/utils/logging_setup.py
create mode 100644 tests/test_logging_setup.py
create mode 100644 tests/test_rally_summary.py
---- verify ----
feat(logging): quieter run.log ? suppress HTTP chatter + end-of-run rally summary
3af7d1a feat(logging): quieter run.log ? suppress HTTP chatter + end-of-run rally summary
```

85. user (2026-06-13T19:58:52.322Z)

File created successfully at: C:\Users\avidu\AppData\Local\Temp\runlog_pr_body.md (file state is current in your context ? no need to Read it back)

86. assistant (2026-06-13T19:58:39.745Z)

Clean message, and `origin/master..HEAD` shows only my single commit. Now push and open the PR (the deliverable calls for a plan in the PR description). Let me write the PR body to a file to avoid quoting pitfalls.

87. assistant (2026-06-13T19:58:52.291Z)

Why

Owner's manual-QA loop: run the tool on a video, then eyeball the rally CSV against `run.log` to judge quality. Two things hurt that today ? the Gemini path floods the log with per-request HTTP lines, and there's no compact rally-times view in the log to line up against the CSV.

Plan ? implementation

****Change 1 ? suppress HTTP chatter (config-gated, default quieter)****

- New central helper `backend/utils/logging\_setup.py::setup\_logging(verbose\_http=False)` : runs `basicConfig` (same format as before) and raises the noisy third-party loggers ? `httpx`, `httpcore`, `urllib3`, `google\_genai`/`google.genai`, `google.auth`, `grpc` ? to `WARNING`. `verbose\_http=True` sets them to `NOTSET` to restore full chatter for request-flow debugging. Our own `backend.\*` loggers are never touched. Idempotent.
- New knob `logging.verbose\_http: bool = False` (`LoggingConfig` on `AppConfig`).
- Wired into the manual-QA run path: `run\_process\_and\_stitch.py` calls `setup\_logging()` at import (quiet default) and again as `setup\_logging(verbose\_http=cfg.logging.verbose\_http)` once config is loaded.

****Change 2 ? end-of-run rally-times summary****

- New pure helper `backend/tools/export.py::format\_rally\_summary(segments) -> str` (sits next to `segments\_to\_csv`/`basic\_stats`): a header (total rallies + total play in `HH:MM:SS`) and a per-rally table `# start end dur(s) shot\_count`, in time order. Sourced from the produced `play\_segments` (the same dicts that feed the rally CSV), so the log lines up row-for-row with the CSV. Handles `metadata` as a dict or a JSON string; missing `shot\_count` renders `.`. Timestamps via a local `\_fmt\_hms` matching the segmenters' `\_fmt\_ts`.
- Logged as one block at the end of `run\_process\_and\_stitch.py::main()` (single timestamp, aligned rows). Also emitted on the all-filtered early-return path so a fully-filtered run still shows what was produced.

Sample output

```
...
=== Rally summary: 3 rallies, 01:00:05 (3605.4s) total play ===
#  start      end      dur(s)  shots
1  00:00:10   00:00:25    15.4     8
2  00:01:35   01:01:20   3585.0    11
3  01:01:40   01:01:45     5.0     -
...
```

Tests

- `tests/test\_logging\_setup.py` ? default raises noisy loggers to `WARNING`; `verbose\_http=True` restores; `backend.\*` untouched.
- `tests/test\_rally\_summary.py` ? header counts/total, `HH:MM:SS` rows, JSON-string metadata parsed, missing `shot\_count` ? `.` , empty list.
- `tests/test\_config.py` ? `logging.verbose\_http` default + file override.

Full suite green (477 passed), `ruff check .` clean, `mypy backend training` clean.

Scope

Config-gated and minimal; default behavior = quieter log + the summary block. Only the manual-QA run path (`run\_process\_and\_stitch.py`) is wired; no unrelated areas touched. Not merging ? owner-gated.

? Generated with [Claude Code](https://claude.com/claude-code)

88. user (2026-06-13T19:58:57.647Z)

```
remote:
remote: Create a pull request for 'feat/runlog-readability' on GitHub by visiting:
remote:   https://github.com/avidullu/badminton-highlight-indexer/pull/new/feat/runlog-readability
remote:
branch 'feat/runlog-readability' set up to track 'origin/feat/runlog-readability'.
To https://github.com/avidullu/badminton-highlight-indexer.git
```

```
* [new branch]      feat/runlog-readability -> feat/runlog-readability
```

89. user (2026-06-13T19:59:07.465Z)

<https://github.com/avidullu/badminton-highlight-indexer/pull/120>

90. assistant (2026-06-13T19:59:16.597Z)

PR opened: **<https://github.com/avidullu/badminton-highlight-indexer/pull/120>**

Per the checkpoint convention, let me update the live current-state memory to record this PR.
First I'll read it to append accurately.

91. user (2026-06-13T19:59:17.797Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-461 of 1667 total (76837 tokens, cap 25000). Call Read with offset=462 limit=461 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-14 (ANALYZERS DOC ? FRAMEWORK + paper framing ? **PR #119 OPEN,
owner-gated, do NOT merge**; branch
13     `docs/analyzers-framework` off master `35a50f3`; docs-only). Owner fed two more ideas
this session and asked to fold them in
14     and treat the doc as a **reusable framework / research paper**, not a project patch
(ties [[paper-coauthor-aim]]). Extended
15     the on-master `docs/ANALYZERS_AND_RECONCILERS.md` (which had landed via #117) from "two
seams" into a **multi-scope
16     signal-fusion framework**: producers emit SIGNALS at **3 temporal scopes** ? frame
detectors (`RallyCue`) / window analyzers
17     / **session analyzers** ? all consumed as feature columns by the learned scorer (spine
unchanged: **no special-casing**, signal
18     ?scorer, never `if/else`). Folded in: (1) **warm-up/practice** as a new
**`SessionAnalyzer`** tier (whole-timeline span ? an
19     `in_warmup` membership feature the scorer DOWN-WEIGHTS, NOT a hard `if t<warmup_end:
drop`; measurable on golden TODAY since
20     warm-up is already labeled negative); (2) **shuttle-state**
(`in_air`/`racket_contact`/`inactive`) placed correctly as a
21     **frame-scope detector** (a MULTI_SIGNAL_FUSION_PLAN `RallyCue`, NOT a window analyzer)
? derive-from-existing-signals first
22     (velocity+visibility+direction-reversals+audio thwack), dedicated per-frame model ONLY
where WASB trajectory gaps bite, with
23     the per-frame-label blocker flagged. Added **$10 generality & research positioning**
(weakly-supervised temporal segmentation;
24     standard-vs-contributed) with an explicit **attribution honesty caveat**: NO fabricated
citations/results, related-work +
25     ablations + >1-sport still owed, n=6 = direction-check only ? the honest backbone the
paper aim needs. Diff = `2-seam?framework`
26     (310+/137?). ? Shared checkout had flipped to master mid-task (the sibling's collision
lesson recurred) ? re-verified branch,
27     branched fresh, normal commit (no in-flight WIP left to isolate; #117/#118 already
merged so all cross-links resolve). PR #116
```

```

28     stays CLOSED (content on master via #117). ?? NEXT (owner-gated): review #119; impl
tracks unchanged (fusion P3 LOVO, then
29     analyzers/reconcilers A1?A5 / R1?R3 per the doc's phasing ? each ONE PR, default OFF,
promote on measured LOVO lift).
30
31     **Checkpoint:** 2026-06-14 (FUSION P2 SHIPPED + collision cleanup + paper aim ? master
`35a50f3`; suite 467 green; 0 open PRs).
32     **PR #117 MERGED = `FusionSegmenter`** (fusion P2, registry `fusion_hybrid`, **default-
OFF**): shuttle-primary (windows seed
33     from WASB/TrackNet) + adjudicating cues ? `net_crossings` (Gate A, always persisted,
GPU-free via rally_gate w/ a reused
34     axis estimate) + optional person cue (Gate B) persisting the ~5 `fusion_features.py`
features (pure-tested) via
35     `PersonDetector.detections_per_frame` over each window's ORIGINAL-video frame range.
Plugs in via **two IDENTITY hooks** on
36     `trajectory_hybrid` (`_enrich_candidate_stats`/`_select_for_handoff`) ? wasb/tracknet
byte-identical (adversarially verified).
37     Served gating optional+default-OFF (`fusion.min_crossings`/`min_players`) ? gates the
handoff while persisting the full
38     SUPERSET (offline substrate intact). Court mask = optional `fusion.court_polygon`.
**Design-first** (`fusion-p2-design`
39     workflow) ? **31-agent adversarial review** (25 findings?19 confirmed; headline = NO-
REGRESSION verified) ? fixes folded
40     (frame-dim guards, round() frame idx, rally_gate `estimate=` reuse, honest torch
docstrings, +6 tests). **PR #118 MERGED** =
41     README index fix (QUALITY_ITERATIONS + EXPERIMENT_HARNESS P1-SHIPPED). **? SHARED-
CHECKOUT COLLISION (lesson):** the
42     spawn_task DESIGN session ran in THIS checkout, committed `0f673ff` (the analyzers
design doc) + flipped the branch BETWEEN
43     my branch-point check and my `checkout -b` ? my fusion branch rode on top ? **#117's
squash absorbed the design doc**, so
44     **`docs/ANALYZERS_AND_RECONCILERS.md` is on master via #117** and **PR #116 was CLOSED
as redundant** (content intact on
45     master; #116's stale "OPEN do-not-merge" checkpoint below is superseded). Re-verify `git
branch` AND that no sibling commit
46     landed since, before `checkout -b`. **Owner side-aim ratified: co-author a paper**
"local, cheap & efficient rally detection
47     via multiple clever ideas" ? [[paper-coauthor-aim]]; `docs/QUALITY_ITERATIONS.md` is the
ablation-ready empirical backbone
48     (honest measured results only). **Analyzers+reconcilers DESIGN on master (owner-gated,
NOT started):** injectable per-window
49     analyzers (e.g. net-hit service-fault) + a report-first reconciliation pass; spine =
signals?learned-scorer, **no
50     special-casing** (owner directive). ?? NEXT (owner-gated): **fusion P3** (person
features ? `experiment.compare` LOVO lift on
51     the 6 golden videos ? the eager-person-compute cleanup lands here) + measure the Gate-A
held-shuttle drop on GPU; then
52     analyzers/reconcilers impl; P4 audio. Earlier this session: PR #114 harness, #115
person-cue extraction (see checkpoints below).
53
54     **Checkpoint:** 2026-06-14 (DESIGN DOC: two extensibility seams ? **PR #116 OPEN, owner-
gated, do NOT merge**; docs-only,
55     branch `docs/analyzers-and-reconcilers` off master `df7d5df`). Produced
`docs/ANALYZERS_AND_RECONCILERS.md` (367 lines, a
56     peer to EXPERIMENT_HARNESS + MULTI_SIGNAL_FUSION_PLAN) + a 1-line `docs/README.md`
cross-link. **Spine (owner directive):
57     NO special-casing in the decision path** ? every analyzer/reconciler emits a SIGNAL
(value+confidence); the SCORING MODEL
58     learns to weight it (already true in miniature: `experiment.predict`/`_gate_mask` treat
`net_crossings`/`players_in_court`
59     as feature columns). **Seam 1 = injectable `WindowAnalyzer`s (forward)**: a registry
(mirrors `segmenter_registry`) of pure
60     per-window cues over the already-produced per-frame-aligned signals (WASB trajectory +
`PersonDetector.detections_per_frame`
61     + `rally_gate` net); each emits an `AnalyzerResult` = confidence-scored features ?

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```

`candidate_segments.stats`, events ?
62     `events` table (buffer-on-candidate, flush-on-confirm), optional boundary hints; worked
example = a **service-fault**
63     analyzer (shuttle halts in net region ? `service_fault` signal + `net_hit` event + END
hint); injects at the FusionSegmenter
64     `_enrich_candidate_stats` hook, flag-gated default OFF. **Seam 2 = reconciliation pass
(backward)**: a report-first "linter
65     for rally decisions" ? `ConsistencyRule`s (trailing-dead-time?truncate *(owner's misfire
fix)*, no-net-crossing?drop,
66     over-merged?split, boundary-slack?tighten) flag decision-vs-evidence conflicts;
**report-first/apply-optional, NEVER silent**;
67     expressible as a harness `Variant` transform so LOVO-measurable on the 6 golden videos
BEFORE auto-apply; idempotent, fixed
68     precedence. ? **DESIGNED AGAINST the IN-FLIGHT fusion-P2 WIP** (uncommitted in the
shared tree: `fusion_hybrid.py`,
69     `fusion_features.py`, `FusionConfig.cue_flags`, `person_detector.detections_per_frame`,
`docs/QUALITY_ITERATIONS.md`) ? that
70     WIP was kept **byte-intact** (I isolated my single README line via git plumbing; tree is
CRLF / autocrlf=true, restored
71     owner README to exact session-start bytes 3841B). **Merge-order:** the doc cross-links
QUALITY_ITERATIONS.md + the fusion
72     hook, neither on master yet ? **merge #116 after (or with) fusion-P2** (noted in the PR
body). ?? NEXT: owner reviews #116;
73     implementation is the phased A1?A3 / R1?R3 plan in the doc (each ONE PR off master,
default OFF, promoted only on measured
74     LOVO lift) ? all owner-gated, none started. Ties [[candidate-segments-finetuning]] + the
harness ([[working-agreement-merging]]).
75
76     **Checkpoint:** 2026-06-13 (RALLY-QUALITY TRACK: HARNESS + PERSON-CUE EXTRACTION SHIPPED
? **PR #114 + #115 MERGED**,
77     master `df7d5df`; suite 421?441 green; 0 open PRs). **PR #115 (fusion P1):** extracted
the torchvision person detector
78     + ByteTrack + velocity windowing out of `yolo_hybrid` into
`backend/pipeline/detectors/person_detector.py`
79     (`PersonDetector`, a reusable cue producer); `yolo_hybrid` now ORCHESTRATES only
(chunk/prefetch/AI-handoff/DB) and
80     delegates Stage-1 to `self.person`. **No behavior change** ? proven by the unchanged
`process_video` output tests (mock
81     seam moved to `self.person`); pure detection unit tests ? new
`tests/test_person_detector.py`. person_detector is
82     mypy-CLEAN (`self.model: Any`) so it needed NO quarantine; the torchvision code that put
yolo_hybrid in quarantine left
83     it, but yolo_hybrid STAYS quarantined for PRE-EXISTING orchestration debt (implicit-
Optional process_video defaults,
84     Success|Failure ABC-override mismatch, a Future[...] annotation ? flagged in mypy.ini to
ratchet separately). ruff+mypy
85     clean, CI 4/4. Docs synced (fusion-plan P1?DONE, CODE_MAP person_detector entry +
yolo_hybrid-as-orchestration, NEXT_STEPS
86     item 2 ticked). ?? **NEXT (owner-steered):** **PR 3 = fusion P2.** The cheap, highest-
confidence, PURE-LOGIC piece =
87     **wire Gate A (`min_crossings`) into the live trajectory path**
(`trajectory_hybrid`/`wasb_hybrid`) + add the knob to
88     `IndexingConfig` ? **fixes the owner's held-shuttle false positive*, default OFF,
immediately A/B-able via the harness on
89     the 6 golden trajectories. Gate B (court-mask + player-count) is **owner-gated** on the
court-mask-source decision
90     (MULTI_SIGNAL_FUSION_PLAN §9.1: manual polygon vs auto homography ? owner). **P3 learned
fusion + the SxS-on-new-videos
91     RUN need GPU/owner-side** (persist person features on real video ? train on the 6 golden
? `compare_unlabeled(CONTROL,
92     +person)` on new footage). Two PRs this session within [[working-agreement-merging]]
self-merge authority.
93
94     **Checkpoint:** 2026-06-13 (EXPERIMENT HARNESS P1 SHIPPED ? **PR #114 MERGED**, master
`ecee786`; suite 421?440 green).

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95     Picked up the rally-quality track (the desktop-agent pick-up `NEXT_STEPS` flagged; both
design docs `EXPERIMENT_HARNESS.md`
96     + `MULTI_SIGNAL_FUSION_PLAN.md` arrived via the session-start `git pull`
15e4e0e?9231af2, PR #112). Built
97     `backend/eval/experiment.py` = the thin variant/experiment layer from
`EXPERIMENT_HARNESS.md` P1 (the audit's "~80%
98     already exists" HELD ? verified file:line; pure glue over
metrics/calibration/classifier/training/regression.gt_hash/
99     gemini_refine.merge_overlaps/query_candidate_segments, **NO new scoring math**):
`Variant` (JSON recipe:
100     segmenter+config_overrides+cue_flags+Gate-A `min_crossings`+Gate-B `min_players`+learned
classifier; `spec_hash`; pinned
101     `CONTROL`), `compare` (labeled A/B, **LOVO-honest** train-on-others for learned
variants, B?A deltas + `label_set_hash`),
102     **`compare_unlabeled`** (the owner's **SxS-on-NEW-videos** ask ? agreed/a_only/b_only
via match_intervals, no GT),
103     `record_experiment` (? `eval_baselines/experiment_leaderboard.json`),
`load_video_candidates` (DB bridge). 19 tests incl.
104     the load-bearing **no-regression invariant** (all cues OFF ? byte-identical to CONTROL);
ruff+mypy clean; CI 4/4. Docs
105     synced (EXPERIMENT_HARNESS P1?DONE, CODE_MAP entry + stale regression-baseline-path fix,
NEXT_STEPS item 3 ticked). Owner
106     answered ramp-up Qs: **corpus = all 6** golden videos (he said "5";
LargeTestVideo/TestLargeVideo may be cuts of one match
107     ? kept all 6 per the manifest); **merge authority = self-merge on green then continue to
PR 2** (within [[working-agreement-merging]]).
108     ?? NEXT (this session): **PR 2 = fusion P1** person/vision-cue extraction (lift
torchvision+ByteTrack+velocity out of
109     `yolo_hybrid` ? `backend/pipeline/detectors/person_detector.py` producer; yolo_hybrid
consumes it, no behavior change ?
110     regression-safe). Then P3 learned fusion trains on the 6 golden videos +
`compare_unlabeled(CONTROL,+person)` SxS on new
111     footage (GPU/owner-side). Tracks [[candidate-segments-finetuning]] + [[collector-
indexer-bridge]].
112
113     **Checkpoint:** 2026-06-13 (WORKING-TREE CLEANUP ? master `15e4e0e`, tree now CLEAN).
Cleaned up the leftover
114     uncommitted bits (the cache-hygiene CODE was already MERGED as #109/#110 per the
checkpoint below ? nothing to rescue
115     there): (1) `config.json` GPU-run knobs **reverted to committed defaults** (was
wasb_hybrid/7200/8 ? gemini/1800/1;
116     values stay recorded in the cache-hygiene checkpoint below ? re-apply
`default_segmenter: wasb_hybrid` in one line if
117     that's the intended new default; user didn't answer keep-vs-revert, took the safe
recoverable path). (2) stray root
118     `arial.ttf` (unused ? shipped watermark font is committed
`backend/assets/fonts/RobotoSlab-Regular.ttf` from #103) ?
119     moved to `scratch/`. (3) genuine untracked draft `docs/STORAGE_SHARING_MODEL.md`
(190-line storage-anchored
120     monetization model, owner co-design pending) ? branched + **PR #111 OPEN (owner-gated,
docs-only ? do NOT auto-merge**);
121     supersedes COMMERCIALIZATION.md framing if ratified ? ADR then). Master tree now shows
ONLY intentional carryovers
122     (untracked `scratch/` handoffs+repro scripts, `artifacts/` Gen-0 bundle) ? could be
gitignored for a pristine
123     `git status` if desired (offered, not done). Reminder: `python -m tools.cleanup_caches
--apply` reclaims the 96.5 GB
124     the SessionStart hook flagged.
125
126     **Checkpoint:** 2026-06-13 (CACHE HYGIENE shipped ? **PR #109 + docs #110 MERGED**,
master `15e4e0e`; + the ?5 reel +
127     manual cleanup). **#110** = diligence docs follow-up (README $Cache Hygiene + knobs +
tools/ line; CODE_MAP detector
128     self-clean, tools/cleanup_caches.py entry, common-task rows; min_shot_count API-parity
note) ? docs-only, CI 4/4 green.

```

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129 Both repos now have **0 open PRs**. After the GX010125 E2E run left ~163 GB in WSL
~/clips`, built a full cache-hygiene system (all on master):
130 (1) **self-cleaning runners** ? `indexing.{wasb,tracknet}.keep_frames` (default false)
deletes the frame cache +
131 staged video after a SUCCESSFUL run; on failure keeps them for resume; (2)
**tools/cleanup_caches.py** dry-run-first
132 ops tool (`--apply`/`--keep-video`/`--older-than`/`--include-wasbcache`/`--summary`);
(3) **disk guard** `min_free_gb`;
133 (4) **SessionStart nudge hook** in `.claude/settings.local.json` (local, gitignored).
Suite 409? **421 green**, ruff+mypy
134 clean. Policy saved ? [[cache-hygiene-policy]]. ? WSL-scan footgun: the owner's login
shell mangles `for e in ` loops
135 (loop var empty) ? use du/find arg-globbing. Also this session (earlier): merged
**#107** (API min_shot_count) + **#108**
136 (watermark Windows colon fix); ran GX010125 E2E on GPU ? **GX010125_highlights.mp4** (11
rallies >7 shots) +
137 **GX010125_highlights_min5.mp4** (24 rallies ?5, watermark now works ? validates #108)
via a stitch-only reuse script
138 (no re-inference); confirmed **Gemini got 1080p** (4K only at final stitch); delivered
an execution flowchart. Manual
139 WSL cleanup freed ~68 GB (kept GX010125's 93 GB frames + proxy per owner choice). ?
LOCAL uncommitted `config.json`:
140 default_segementer=wasb_hybrid / min_shot_count=8 / wasb.timeout_sec=7200 from the run.
See [[gotcha-long-runs-gpu-wsl]].
141
142 **Checkpoint:** 2026-06-13 (SYNC + HANDOFF REFRESH ? read-only reconciliation by a ramp-
up session; no new code from
143 me this pass). `git pull --ff-only` = already up to date at **master `6de70aa` (#108)**.
Reconciled remote vs my last
144 session (#100): seven PRs landed from OTHER slates (NOT my work ? facts from git/gh):
**#101** configurable branding
145 watermark . **#102** thread config stitching paddings into live VideoCompiler sites .
**#103** bundle Apache-2.0
146 default watermark font + fallback log . **#104** C13 field kit (flyer/playbook/answer-
sheet/signals) . **#105** i18n
147 design plan (Kannada-first) . **#107** enforce `stitching.min_shot_count` on
/api/compile . **#108** escape Windows
148 drive-letter colon in ffmpeg filtergraph (fixes #103 watermark on Windows). **OPEN:
#106** docs/video-locality-model
149 (awaiting owner). ? **UNCOMMITTED WIP IN TREE (sibling/owner ? DO NOT CLOBBER):** a WSL
**cache-hygiene** slice in
150 progress, built on my `WslCommandMixin` ? `base.py` (`_wsl_free_gb`/`_wsl_rm_rf`),
`wasb_runner.py`+`tracknet_runner.py`
151 (`keep_frames`/`min_free_gb` config + pre-run disk guard + post-success cache delete),
`models.py` (new fields),
152 `config.json` (the GPU-run knobs: default_segementer=wasb_hybrid, min_shot_count=8,
wasb.timeout_sec=7200), untracked
153 `arial.ttf` (watermark font) + `docs/STORAGE_SHARING_MODEL.md`. Not yet PR'd by whoever
owns it. [[session-handoff]]
154 rewritten to this state.
155
156 **Checkpoint:** 2026-06-13 (FIRST FULL E2E GPU RUN on a real 4K GoPro match + 2 merged
fixes).
157 Owner asked to run the indexer E2E on the GPU over `F:\GoPro Hero Black
12\KhelSutra\Badminton 12
158 June 2026\GX010125.MP4` (12.8-min 4K60, 11.5GB), highlights for rallies with >7 shots,
config-driven.
159 - **DELIVERED:** `output\GX010125_highlights.mp4` ? **11 rallies (shot_count 8?16) of 52
detected**;
160 true 4K 3840x2160 h264 59.94fps, 2:23, 1.15GB; stitched from the 4K ORIGINAL (proxy
only for indexing).
161 wasb_hybrid: 64.8% shuttle visibility, 18 candidate windows ? 52 Gemini-confirmed
rallies. min_shot_count=8
162 correctly excluded a 7-shot rally (">7" semantics). Gemini ran clean (no 503s);
~47-min wall clock.

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163      - **2 PRs merged on green CI** (owner's [[working-agreement-merging]] standing
authorization):
164      **#107** `/api/compile` now enforces `stitching.min_shot_count` (parity w/
run_process_and_stitch);
165      **#108** stitcher `_ff_quote` escapes the Windows drive-letter colon ? fixes the on-
by-default
166      bundled-font watermark (PR #103) silently failing on Windows (`fontfile='C:/...'`
broke ffmpeg
167      drawtext parsing; also fixed absolute logo `movie=` paths). Both with regression
tests; suite 409 green.
168      - **The hard-won run recipe is in [[gotcha-long-runs-gpu-wsl]]:** Modern Standby killed
the first 2
169      overnight attempts (keep-awake fix); cv2 PNG extraction (~38min) is the real
bottleneck not the GPU
170      (ffmpeg pre-extract, count-exact for CFR ? skips it); 1080p-proxy-index / 4K-stitch;
per-phase
171      observability (cache manifest for WASB, handoff dir for Gemini). Reusable scripts in
`scratch/`
172      (run_gx010125.py, keep_awake.ps1, probe_.py).
173      - ? LOCAL UNCOMMITTED `config.json`: default_segmenter=wasb_hybrid, min_shot_count=8,
wasb.timeout_sec=7200
174      (set for this run; revert to gemini/1/1800 if not wanted). Watermark fix NOT applied
to the delivered
175      reel (it ran before #108); a re-stitch would hit the DB segment cache + now-working
watermark (~5min).
176
177      **Checkpoint:** 2026-06-12 (latest ? PAID COMPONENT REMOVED + VIDEO-LOCALITY MODEL; **PR
#104 updated `9c84fea` +
178      PR #106 OPEN `85abe26`, both CI green, both AWAITING OWNER**). Owner spec: (1) remove
ALL associate-pay from the
179      field kit ("something else in mind") ? flyer/playbook now "Engagement terms: discussed
individually"; honesty
180      mechanics reworded ("judged on honest capture, not positive answers"); KEPT: WTP
research questions (study subject)
181      + the ACADEMY's Phase-2 session fee (different party ? flagged, owner may strip too).
(2) Storage/upload questions
182      answered from code: uploads ARE stored in full (`security.upload_dir`, NO retention
policy ? BACKLOG); 10GB CANNOT
183      upload (max_upload_mb=4096 rejects at init; ?90MB sequential parts; hours on Indian
uplinks). (3) NEW
184      `docs/VIDEO_LOCALITY_MODEL.md` (#106): central insight = timestamps are the product,
video is dead weight (only the
185      compiler touches original bytes, cuttable where the original lives); locality ladder
L0?L4 (L2 proxy-upload reuses
186      the collector ADR-002 frame-identical contract; L1-direct = client?Gemini, zero bytes
through us; L0 = owned-model
187      north star; pilot = L4 sneakernet); **frugality frame (owner): bills affordable iff GPU
doesn't burn ? L2/L1 serving
188      is GPU-less by construction, Gemini €/hr, GPU stays on the owned 2070S, per-video cost
ceiling required**
189      **academy-visit dividend: Phase-2 consented recordings ARE the multi-court target
corpus, show-back = live L2 dress
190      rehearsal, demo reactions = build backlog**
191      10 open questions OQ1?OQ10 (retention, MD5
identity, Gemini consent,
192      key custody, packaging, review source, reel delivery, uplink data, volume, L3 survival);
rec lean = L2 default
193      (NOT a greenlight). NB: #106's base commit was pushed by owner/sibling mid-session
(shared checkout). ?? owner:
194      review/merge #104 + #106; decide academy-fee keep/strip; the "something else" comp
structure is his to define.
195
196      **Checkpoint:** 2026-06-12 (latest ? C13 KIT REFRAMED PRODUCT-FEEDBACK-FIRST per owner
spec + **RECRUITING FLYER
197      added; PR #104 UPDATED `985a329`, CI 4/4 green, AWAITING OWNER REVIEW** ? do NOT merge
without him). Owner spec:

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197 hire someone to visit academies, talk to coaches AND students (how does it help +
 improvement feedback); stretch =
 198 record ? pipeline ? show the academy ITS OWN sample result; UI/hosting "ready by then".
 Kit now = 4 docs by audience:
 199 ****`PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md`**** (NEW ? candidate-facing recruiting flyer, owner
 fills ?/phone + recruits) .
 200 ****`FIELD\_VISIT\_PLAYBOOK.md`**** (RENAMED from FIELD_RESEARCH_ASSOCIATE_FLYER ? day-one
 playbook: demo-after-Q6 rule,
 201 adult-18+-students-only w/ coach-points-out gating, Phase-2 show-back + scripted "can we
 keep using it?" no-promise
 202 reply) . answer sheet (+demo-reaction/student blocks) . PMF_FIELD_SIGNALS (verdict
 machinery UNCHANGED; demo/student
 203 data = product feedback only). 2nd adversarial critique (`w5aexakz2`, 30 agents): 12
 confirmed+fixed (key honesty
 204 fixes: "web app IS opening"? "we're building?will be able to try"; "match report"? "simple
 summary"; "finds EVERY
 205 rally" dropped; minors loophole in "senior/adult students" closed), 15 refuted. ??
 owner: review #104 (thresholds
 206 \$3 + flyer copy) ? merge ? fill ?/phone ? trim 60s demo reel ? recruit.
 207 Owner ratified in-chat: ****PMF validation = physically visiting Bangalore academies and
 reading signals.**** Built the
 208 decision layer on top of the existing `docs/FIELD\_RESEARCH\_ASSOCIATE\_FLYER.md` (which IS
 the "recruit hand-out" the
 209 owner remembered; the khelacharya PDFs in OneDrive are the separate internship flyer ?
 see
 210 **[[sibling-internship-repo]]: ****`docs/PMF\_FIELD\_SIGNALS.md`****** (H1-H7 ? script-question
 map, G/A/R anchors, H5
 211 counting rule = pre-ladder-or-budget-converted ??300 only ? anchored ?300 NEVER counts,
 pinned stopping rule n=15
 212 target/logistics-only/verdict-once, numeric verdicts PROCEED / PIVOT A record+index /
 PIVOT B partner-API
 213 (PB VisionxPodPlay footnote now in-repo) / PIVOT C parent-payer / FALSIFIED) +
****`docs/FIELD\_VISIT\_ANSWER\_SHEET.md`****
 214 (printable per-visit form: verbatim answers, required most-negative quote, before/after-
 ladder tick, OFFICE-USE
 215 score grid) + flyer Q5 rewritten open-ended-first (old wording read the ?100/300/500
 ladder ALOUD ? center-bias
 216 manufactures the exact ?300 PROCEED threshold; caught by the 38-agent adversarial
 critique `ww0rvaqf3`: 1 high/2 med/
 217 2 low confirmed+fixed, 9 refuted) + Stupa added to listen-for + NEXT_STEPS item 5 = kit
 COMPLETE. ?? owner: review
 218 #104 thresholds ? merge ? fill flyer ?/phone placeholders ? recruit the associate ?
 first 2 visits paired. PDFs of
 219 flyer+answer-sheet: generate AFTER placeholders are filled (offer stands).
 220
 221 ****Checkpoint:**** 2026-06-12 (latest ? ROORA VENDOR LOOP COMPLETE: ****collector #23+#24+#25
 MERGED**** sequentially on
 222 green CI, owner pre-authorized; master `ca61405`, 0 open PRs, only owner's `.gitignore`
 edit uncommitted).
 223 ****#23 boundary_grid_snap BLOCKER**** in detect_issues (gcd-of-boundary-frame-diffs > ±0.3s
 bar ? gate BLOCKED; skips
 224 CSV-path/no-fps/<3 rallies) ? ? ****replayed against the REAL 6-June PB+TT pilot zips:
 both auto-block with "All 12
 225 rally boundaries sit on a 30-frame grid (~0.50s @ 59.94fps)"**** (the manual forensics,
 now automated) + ROORA_BRIEF
 226 Phase-B revision (\$7 REQUIRED method: CVAT step 1, first/last box on exact contact/dead
 frame, sparse interior OK;
 227 \$4 grid-snap auto-detected warning; \$2 skip-rallies-in-progress-at-file-edges; \$1
 annotation-optimized-copy note).
 228 ****#24 prep-annotation-batch****: mirrors `//` ? proxies-
 only tree w/ same layout
 229 (thin orchestrator over the hardened single flow; per-video isolation; resumable "have"
 skips; live 2-court E2E
 230 smoke green). ****#25 ROORA_RUNBOOK.md**** (docs/annotation/): operator E2E ? send checklist
 (layout?batch?manifest

231 commit?intake+email w/ the 2 non-negotiables), receive (gate decision table incl. grid-snap, review sheet,
232 import/export on ORIGINALS), troubleshooting table for every guard. Suite **180 green**.
?? Vendor loop is now fully
233 tooled+documented. ?? AGREED NEXT (owner, session wrap 2026-06-12): (1) owner fills the canonical doc's TODO(avidullu)
234 markers + screenshots, labels it **v8**; (2) **joint TEST RUN of the full vendor loop on real footage BEFORE
235 anything goes to Roora** (runbook §1 send-side end-to-end: batch layout ? prep-annotation-batch ? manifest ?
236 mock-receive via validate-delivery on an old delivery); (3) only then send. **v7 ANOMALY RESOLVED (2026-06-12).**
237 the canonical vendor-facing Brief+Guide is the Google Doc at docs.google.com/document/d/1NvAXE0q7t4Z0QddLkEfnta0DU982yDZaqkGNPMe8EY0
238 (provenance: generated from master `ca61405`; supersedes Drive v3/v5/v6/v7; owner confirmed superset-of-v7) ? see
239 [[roora-guidelines-drive-doc]]; before sending: pick ONE version label + fill its TODO(avidullu) markers/screenshots.
240
241 **Checkpoint:** 2026-06-12 (earlier ? ADR-002 IMPLEMENTED: **collector #22 MERGED** `a068fd1`, owner authorized
242 merge-on-green; #21 ADR + #20 + indexer #102/#103 all merged earlier today, slate was clean). Shipped: `curate
243 prep-annotation-proxy` + `src/core/proxy.py` ? ?1080p/-g 15`/CRF 19/audio-kept re-encode w/ `--fps\_mode passthrough`
244 (NEVER `-r`), post-encode parity verify (exact frame count; fps $\pm 0.1\%$; duration like-with-like stream/container),
245 delete-on-ANY-failure (incl. Ctrl-C), VFR refusal w/o `--allow-vfr`, provenance ?
246 `data/annotation\_proxy\_manifest.csv` (source+proxy MD5s; video_id collision guard BEFORE encode; Excel-lock prints
247 row + keeps proxy), same-file guard normcase/realpath/samefile (NTFS case-variant `--out`+`--force` would have let
248 `ffmpeg -y` TRUNCATE the source ? review verifier reproduced it live pre-fix). Suite **169 green** (+32), CI green.
249 Verified live on ffmpeg 8.1.1 (smoke clip 59.94fps: parity 179 frames both files; smoke caught `--vsync` deprecation ?
250 `--fps\_mode`). Process: ultracode adversarial review workflow (`wyv8524zf`, 18 agents) confirmed 1 high/2 med/7 low ?
251 all fixed pre-merge. Docs: INGESTION_PIPELINE step 0 + diagram leg + onboarding FAQ + sampling-FAQ split
252 (highlights-vs-golden), ADR-002 provenance bullet as-implemented amendment, PIPELINE_ROLE ADR refs qualified.
253 ?? REMAINING ADR-002 follow-ups (separate items, NOT started): Phase-B brief revision (frame step 1 / exact boundary
254 frames / "we supply proxies") ? owner words vendor comms; grid-snap QA check in validate-delivery. First real use:
255 run prep-annotation-proxy on the next Roora batch before handoff.
256
257 **Checkpoint:** 2026-06-12 (earlier ? ANNOTATION-PROXY DECISION RATIFIED ? **collector PR #21 OPEN**, docs-only
258 **ADR-002** in collector `docs/DECISIONS.md`). Owner agreed in-chat w/ my rec from the assessment checkpoint below:
259 proxies generated in-repo via a future `curate prep-annotation-proxy`. ADR records the proxy contract (SAME fps ?
260 never reduce; no trim; frame-count parity check; 1080p/CRF18-20/-g15/audio kept; intake-manifest row w/ original+proxy
261 MD5s), the why-collector rationale (owns the vendor flow = command #0 of convert-cvat?validate-delivery?import-local;
262 verification+provenance is the durable 80%, one-off scripts skip it), and the **Phase-B necessity argument the owner
263 wanted captured: ~12-15 videos/sport $\times$ ~30 min 4K/59.94 = >100k frames/file ? vendor frame-step-1 annotation
264 infeasible without proxies ? near-precondition, not optimization**. Follow-ups recorded IN the ADR: Phase-B brief
265 convention line (step=1, first/last box on exact contact/dead frame) + re-add grid-snap

QA to validate-delivery.

266 Collector checkout back on master (owner's stray `.gitignore` edit untouched). ? sibling
PR #20 (below) edits

267 rally_cvat.py docstring rationale 0.5/1.0?0.5/0.5 ? no file overlap w/ #21, both docs-
true. ?? owner reviews/merges

268 #21 (+ #102/#20 below); subcommand IMPLEMENTATION stays owner-gated (separate PR on
greenlight).

269

270 ****Checkpoint:**** 2026-06-12 (stitching-padding wiring fix ? ****indexer #102 + collector**
#20 OPEN, CI green, awaiting
271 owner review**). Audit finding (d) (2026-06-10) fixed: config.json
`stitching.{begin,end}\_padding` was silently
272 ignored by the live compiler ? both `backend/main.py` VideoCompiler sites
(`/api/compile` + CLI trim) passed only
273 `watermark=` , so reels always used ctor defaults 0.5/****1.0**** vs configured 0.5/0.5. Fix
threads the typed
274 StitchingConfig paddings into both sites (+ `StitchingConfig` re-export from
backend.config); regression test drives
275 the REAL compiler through /api/compile (mocked ffmpeg) asserting a config override
(2.0/3.0) reaches `-ss/-to`
276 (8.0/17.0 for a 10?14s segment ? unwired defaults would give 9.5/15.0); CODE_MAP truth-
sync'd (ctor-vs-config
277 distinction + recipe row now cites the main.py wiring). ? ****Behavior change on merge:**
live API reels end padding
278 1.0?0.5s****** (the config value). Companion ****collector #20**** = docstring-only:
`rally\_cvat.py::detect\_issues` cited the
279 hardcoded 0.5/1.0 + a stale module path; now cites config-driven 0.5/0.5 (rationale
still holds; ? cross-checks the
280 sibling Roora-proxy thread below ? boundary-slack rationale now assumes 0.5s end
padding, not 1.0s). Verified: indexer
281 suite ****401 green**** + 4/4 CI lanes; collector ****137 green**** + CI. Both repos back on
master; owner's uncommitted
282 collector `.gitignore` edit untouched. NOTE: untracked `arial.ttf` at indexer root (not
mine ? left alone, likely a
283 watermark-font experiment). ?? owner reviews/merges #102 + collector #20.
284

285 ****Checkpoint:**** 2026-06-12 (Roora ANNOTATION-PROXY assessment delivered ? AWAITING OWNER
DECISION, no code).

286 Owner asked: re-encode videos to a Roora-friendly profile before handoff? Researched via
workflow `wr6mnel6c`
287 (3 sweeps + adversarial verify; full JSON
`%LOCALAPPDATA%\Temp\claude?\tasks\wr6mnel6c.output`). ****Facts:**** root
288 cause of the ?0.5s boundaries = 30-frame CVAT sampling grid (NOT playback delay;
direction never measured ? docs say
289 "off", not "late"); same grid was flagged [HIGH] in the BADMINTON round too
(`data/roora\_badminton\_issues.txt`) before
290 PB/TT; operative vendor contract = CVAT XML, times recomputed `frame/fps`
(`rally\_cvat.py:266`), collector joins by
291 human video_id NOT MD5 ? proxy handoff is safe if timeline-preserving. ****My rec:**** add
`curate prep-annotation-proxy`
292 to the COLLECTOR (not a loose script): 4K?1080p, SAME fps (never drop ? their 30-frame
habit at 30fps = 1.0s grid),
293 keep audio, dense keyframes (-g 15), frame-count parity check, manifest row w/
original_md5+proxy_md5. Proxy alone ?
294 fix ? must pair w/ the review's "first/last box on exact frame, step=1" convention in
the Phase-B brief. Also: re-add
295 a grid-snap QA check to validate-delivery (detect_issues deliberately dropped
coarse_sampling, rationale =
296 highlight-padding absorption ? conflicts w/ golden $\pm 0.3s$ need). Spawned chip
`task\_d8df38bc` (stitcher padding config
297 not wired into VideoCompiler ? audit finding (d) re-verified live at main.py:508/633).
?? owner decides: collector PR
298 vs one-off; then draft PR on greenlight.
299

300 ****Checkpoint:**** 2026-06-11 (UI-alpha ****PR-2 #99 MERGED + docs-sync #100 MERGED**** ? owner

authorized merge w/ due
301 diligence; master `020ff61`). **LIVE E2E AT MERGE (server up on :8123, real
artifacts):** 86MB smoke video uploaded in
302 3 sequential parts ? reassembled to **MD5 byte-identity** with the source; negatives
verified (out-of-order 409,
303 bad-ext/traversal 400, unknown-id 404; download traversal blocked at BOTH layers ?
handler 400 on backslash form,
304 router 404 on %2F form); **compile of the real confirmed rally ? 3.9MB reel streamed via
GET /api/highlights** (the
305 fixed segment_ids path works browser-equivalent). #100 = doc due diligence: NEXT_STEPS
(PR-2 merged + sweep recorded),
306 UI plan PR-2 checked off w/ E2E evidence, **BACKLOG shlex item closed** (residual
tenancy note kept). The CUJ runbook
307 (PR2_HANDOFF \$"Tomorrow-morning") is now fully unblocked for the owner. ?? NEXT: owner
CUJ; PR-3 identity (remember:
308 `allowed\_video\_root` must contain `upload\_dir`). Original handoff context below:
309 (superseded) **Checkpoint:** 2026-06-11 (UI-alpha PR-2 executed ? #99 was awaiting
review ? per the Cowork handoff
310 `scratch/PR2\_HANDOFF.md`; protocol = Cowork designs, Claude Code runs mechanics, **Avi
reviews/merges ? do NOT
311 auto-merge #99**). #88 confirmed CLOSED ? PR88_FIX skipped. Cowork's uncommitted B2+B3
implementation verified to
312 layer cleanly on post-sweep master (AI-7 CODE_MAP entries + pydantic fixes intact) ?
Step 0 run here (Cowork sandbox
313 down): suite **388 green** (15 new `tests/test\_upload\_download.py`), mypy clean; 3
trivial ruff B904s in the new
314 endpoints fixed per the handoff's trivial-fix rule (exception chaining only, no design
changes). Branch
315 `feat/upload-download`, exactly the 8 handoff files committed ? **including the two
long-preserved doc edits**
316 (DEFERRED_HARDENING_PLAN #16-banner + UI_ALPHA_EXECUTION_PLAN working-agreement
rewrite), which are now COMMITTED
317 (tree finally clean of them). CI 4/4 green on #99. ?? NEXT: owner reviews/merges #99 ?
then the tomorrow-morning CUJ
318 runbook in the handoff (record ? localhost:8000 ? path-paste ? index ? compile ? ?
download; gemini default,
319 rotated key in .env, ~\$0.05/run). PR-3 = per-user dirs + allowed_video_root?upload_dir
(hosted-mode note).
320
321 **Checkpoint:** 2026-06-11 (8?9 QUALITY SWEEP ? COMPLETE ? all approved AIs landed;
master `31069cb`). Final stretch:
322 **AI-7 hybrid-twin collapse** = shared `segmenters/trajectory\_hybrid.py` engine
(tracknet/wasb now ~40-line subclasses;
323 one `\_shot\_count\_from`; `\_wsl\_bash` ? `detectors/base.py::WslCommandMixin`;
DetectorRunner ABC now declares the
324 parse/windowing contract; mypy quarantine ratcheted ?2; 13 equivalence tests; CODE_MAP
2.1/2.2 updated). ? PROCESS SLIP
325 (disclosed): AI-7 commit `2c241a8` went DIRECTLY to master (post-#96 script left
checkout on master) ? content was
326 pre-approved + locally verified; master CI green post-hoc; chose NOT to force-rewind
(concurrent sessions live).
327 - **QUALITY CHECKPOINT (owner-mandated) PASSED:** offline LOVO reproduced **EXACTLY
0.626** (Mahadevpura Gen-0 0.744)
328 post-sweep ? featurizer/eval edits numerically inert. **Live smoke = wasb_hybrid's
FIRST-EVER verified E2E run**
329 (3-min Mahadevpura slice): P1 healthcheck?P2 trajectory (9 candidates, correct
video_id keying)?P3 Gemini (503
330 capacity wave; 6-attempt backoff worked; 1 segment confirmed)?P4 add_play_segment +
candidate link
331 (`wasb-hybrid-gemini-flash-latest`). 1/9 confirms = Gemini weather, NOT code.
332 - ? Smoke found 2 REAL pre-existing live-path bugs (masked: trajectories were always
hand-scripted, tests mock WSL) ?
333 **#97 MERGED**: (1) shlex.quote strangled `~` expansion in staging (the BACKLOG
"shlex-quote breaks ~" item ? CLOSED;
334 fix = `wsl\_tilde\_quote`, quotes only after `~/`); (2) relative `output/wasb` resolved

against WSL cwd ? 8-min GPU run

335 wrote CSV inside WSL invisibly (fix = abspath before /mnt translation). Regression tests pin both command shapes.

336 - Merged this sweep: ****#90** hygiene · **#91** pydantic model_dump · **#92** RUFF lane (+194 fixes; DELETED**

337 never-compiled scripts/generate_khelacharya_flyer.py ? line 39 was a literal AI-placeholder sentence) · ****#93** MYPY**

338 lane (7-module quarantine w/ ratchet rule) · ****#94** SQLite conn-leak fix (340?4 ResourceWarnings) · **#95** coverage**

339 floor 70 (CI-measured 72%) · ****#96** config unknown-key warning · **#97** WSL tilde/abspath · (**#98** Node-24**

340 actions bump = owner-run chip session). Suite ****373 green****; CI = 4 lanes (smoke/lint/mypy/full+cov-floor).

341 - NEW session gotchas: `Select-Object -Last N` on a background pipe loses EVERYTHING if the process is killed ? pipe

342 long background runs through `Out-File` instead; ****sibling sessions flip branches in THIS shared checkout**** (the

343 node24 chip did, mid-session) ? re-verify `git branch --show-current` before every commit; idle/suspend KILLS

344 background runs silently (smoke run #3 died mid-Gemini, no notification, stale report on disk).

345 - ?? NEXT: unchanged owner-gated tracks (UI-alpha PR-2 upload/download per `docs/UI\_ALPHA\_EXECUTION\_PLAN.md`; corpus

346 growth; ship-gate target). Deferred-with-intent from the sweep: ruff I/UP rule groups; yolo_hybrid P3/P4 fold-in;

347 pyproject packaging (AI-8 ? owner deselected); extra='forbid' flip (awaits gemini_* alias retirement).

348

349 ****Checkpoint:**** 2026-06-11 (CI Node-24 action bump ? ****#98 MERGED****, master `7717e15`). GitHub forces actions onto

350 Node 24 from 2026-06-16 (Node 20 removed from runners 2026-09-16); ci.yml pinned checkout@v4 + setup-python@v5 (Node 20).

351 Verified latest majors via `gh api releases/latest`: ****checkout v6**** (v6.0.3 ? NOT v5 as the run-log warning suggested)

352 + ****setup-python v6**** (v6.2.0). Bumped both in all 4 lanes; all green on the PR (lint 11s / mypy 29s / smoke 15s /

353 full+cov 1m27s), squash-merged on verified rollup. PR #97 (fix/wsl-tilde-quoting) was open + untouched throughout;

354 working tree restored to it (PR-2 doc edits intact).

355

356 ****Checkpoint:**** 2026-06-11 (8?9 QUALITY SWEEP IN PROGRESS ? owner approved 8 AIs via question UI; AI-8 packaging NOT

357 selected). ****MERGED** on green CI: **#90**** (hygiene: .coveragerc repointed post-#40 + TEST_RUN_SUMMARY ? docs/archives),

358 ****#91**** (pydantic .dict()?model_dump, 7 sites, warnings 15?0), ****#92**** (? RUFF LANE: ruff.toml E/F/W/B py38 +

359 fix 194 violations + `lint` CI job; ****DELETED** scripts/generate_khelacharya_flyer.py ? never compiled, line 39 was a

360 literal AI-placeholder sentence**, invisible because nothing swept scripts/), ****#93**** (MYPY LANE: mypy.ini lenient

361 green baseline, explicit 7-module quarantine w/ ratchet rule ? main.py+5 segmenters+registry; fixed 18 annotation-level

362 errors), ****#94**** (? DB FIX surfaced by coverage: `with sqlite3.connect()` = transaction scope NOT close ? all 24

363 Database call sites leaked connections; new `\_connect()` CM; ResourceWarnings 340?4), ****#95**** (COVERAGE FLOOR:

364 full-suite lane runs --cov, ****--cov-fail-under=70**** vs CI-measured 72% ? ratchet-only), ****#96**** (config unknown-key

365 warning: dotted-path [CONFIG WARNING] for typo'd/dropped keys at all nesting levels; repo config.json verified clean;

366 extra='forbid' still deferred). Suite 349?****351 green****. CI = 4 lanes (smoke/lint/mypy/full+cov). ?? NEXT: ****AI-7**

367 hybrid-twin collapse** (shared trajectory-hybrid base, fix shot_count drift, dedupe _wsl_bash, equivalence tests) +

368 ****post-AI-7 quality-eval checkpoint**** (owner: verify ~0.66-class evals stick; GPU free,

Gemini key ROTATED, live calls
369 OK). Deferred-with-intent this sweep: ruff I/UP rule groups; yolo_hybrid P3/P4 fold-in (noted as follow-up).
370
371 ****Checkpoint:**** 2026-06-11 (PR #88 ? ****#89 MERGED**** ? KhelSutra UI-alpha docs bundle on master `34cf26c`). #88
372 conflicted (branch cut from `feat/async-job-model` pre-squash of #87) ? per
373 `scratch/PR88\_FIX.md`, recreated as
374 `docs/khelsutra-plan-2` off post-#87 master keeping 6 paths (UI_PROPOSAL, HOSTING_PLAN, UI_ALPHA_EXECUTION_PLAN,
375 mockups/, DECISIONS ****ADR-009****, NEXT_STEPS sync); #88 closed + branch deleted. CI
376 green-gated (statusCheckRollup),
377 squash-merged.
378 - ****Dropped from #89, preserved for PR-2 as UNCOMMITTED working-tree edits on master:****
(1)
379 `docs/DEFERRED\_HARDENING\_PLAN.md` "#16 DONE ? MERGED (PR #87)" status hunk (also saved
as
380 `scratch/PR2\_deferred\_hardening\_status.patch`); (2) `docs/UI\_ALPHA\_EXECUTION\_PLAN.md`
working-agreement rewrite
381 (Cowork designs + leaves `scratch/PRn\_HANDOFF.md`; ****Claude Code executes all git/PR mechanics**** ? ratified
382 2026-06-11). Don't `checkout --` these away; they commit with PR-2.
383 - ?? Next UI-alpha slice = ****PR-2 (upload/download)**** per
384 `docs/UI\_ALPHA\_EXECUTION\_PLAN.md`.
385 - ? ****SLATE CLEANED (2026-06-11):**** indexer = NO open PRs, master==origin (`34cf26c`),
****all 23 stale local branches**
386 pruned** (every one mapped to a MERGED PR #62?#87 before `-D`); only `master` remains.
Collector = NO open PRs, on
387 master. Intentional carryovers (NOT mess): the 2 uncommitted PR-2 doc edits (above),
388 untracked `scratch/` handoffs,
389 untracked `artifacts/rally\_tal\_gen0/0.1.0/` (Gen-0 bundle, ADR-008). ?? Collector has
a stray uncommitted
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****Checkpoint:**** 2026-06-10 (GEMINI BATTERY COMPLETE ? full coverage; owner's billing was fine, postpay active; ALL failures were burst/capacity artifacts). Merged this stretch: ****#80**** (chunker re-encode/downscale, from the chip session ? reviewed+merged), ****#81**** (rescued the chip branch's orphaned run_signals test via cherry-pick), ****#83**** (provider: 6 generate attempts exponential backoff+jitter ? saved chunks live), ****#84**** (tracked
`eval\_baselines/gemini\_wrapper\_2026-06-10.json`). Suite 336 green.
- ****FINAL QUALITY TABLE (IoU .5 vs human golden labels):****
- ****Mahadevpura**** (clean-ish, 31 rallies): heuristic 0.657 · Gen-0 0.744 · ****two-stage WASB+Gemini refine 0.83****
(P .81 R .84, 32 preds) · ****full-video wrapper 0.93**** (P 1.0 R .87).
- ****Kushagra**** (multi-court, 32 rallies, FULL 7/7 coverage + audio): heuristic 0.157 · Gen-0 0.561 ·
****wrapper 0.66**** (P .69 R .62, ±1.2s; FPS = background-game pickups + one 53s dead-time merge ? exactly the contrast-metric confound).
- ****STRATEGY (refined):**** clean footage commoditized (0.93 cheap); ****multi-court drops the wrapper to 0.66 = the owned model's ship target**** (Gen-0 is 0.10 behind at n=6 ? within corpus-growth reach). Two-stage refine quality (0.83) < full wrapper on short videos; its value = COST on long academy tapes (only candidate clips uploaded).
Wrapper reliability is genuinely flaky (503 waves all session) ? provider-fallback architecture is load-bearing.
- Artifacts: `output/mahadevpura\_gemini\_refined.csv`,
`output/MahadevpuraSingles\_highlights.mp4` (demo reel),
Kushagra preds under the ORIGINAL corpus MD5 now. Total Gemini spend today: ~\$1-2.

```

406 - PENDING/owner: rotate the chat-pasted key; ship-gate F1@tIoU decision (multi-court
target now has a number: 0.66);
407 C13 interviews + camera pilot; repo rename.
408
409 **Checkpoint:** 2026-06-10 (GEMINI BASELINE LANDED ? owner's "gradual ramp-up" theory
CORRECT). New key in gitignored
410 `.env` (rotate later ? pasted in chat). The "credits depleted" 429 was a BURST-THROTTLE
artifact: text ping + serial
411 upload worked ? **PR #79 merged**: gemini segmenter now honors `indexing.ai_max_workers`
(was hardcoded 5; config ships
412 1=serial ? at 5 workers the same video 429'd, at 1 it worked).
413 - ? **WRAPPER BASELINE (gemini-flash-latest, ~$0.2-0.5 total): Mahadevpura F1 0.93** (P
1.00 R 0.87, mIoU 0.80, ±0.7s;
414 3 of 4 misses = short rallies Gemini FOUND but our min_segment_duration=3.0 rejected)
vs Gen-0 learned 0.744 vs
415 heuristic 0.657. **Clean footage = commoditized.**
416 - ?? **MOAT TEST (Kushagra multi-court, windowed n=12 GT): wrapper F1 0.56** (P .54 R
.58 mIoU .69) ? Gen-0 learned
417 0.561, both >> heuristic 0.157. **The multi-court confound hurts Gemini too ? the
wedge survives on the target
418 footage class.** Caveats: only chunks 2+4 answered (5/7 lost to GENUINE gemini-flash-
latest capacity 503s ? retry
419 off-peak for full coverage); visual-only (audio stripped in downscale); 1280p re-
encode (4K chunks were 1.3GB >
420 500MB upload cap ? the audit's silent-oversize-skip fired live; fix = chip
task_44e72c41 ? see PR #80 checkpoint
421 below). Kushagra preds keyed under downscaled-file MD5 (`output/kushagra_1280.mp4`).
422 - ? **DEMO ARTIFACT: `output/MahadevpuraSingles_highlights.mp4`** ? real 27-rally reel
compiled by VideoCompiler
423 (the once-broken module) from Gemini segments. The academy-pitch object exists.
424 - **STRATEGY READ:** product path = Gemini for clean footage TODAY (cheap, F1 0.93) +
owned model's job = multi-court
425 (0.56 is the bar to beat there, not 0.93); 503 flakiness = real reliability cost of
the wrapper (provider-fallback
426 story strengthens). PENDING: gemini_refine two-stage (503-gated, run off-peak), full
Kushagra re-run, PR #80 merge.
427
428 **Checkpoint:** 2026-06-10 (4K gemini oversize fix ? **MERGED: #80 + #81 + #82; master
`f01387e`, CI rollup SUCCESS**).
429 Root cause of the live "0 segments / success" on KushagraYashAviFirstVideo.MP4 (4K):
gemini segmenter stream-copies 90s
430 chunks ? all 7 at 832?1299MB > provider `MAX_UPLOAD_MB=500` ? every chunk silently
skipped. Fix (3 layers): (1) chunks
431 RE-ENCODED + downscaled to new declared knob `indexing.ai_chunk_scale_width` (default
1280 = gemini_refine's
432 `--clip-scale-width`; 0 = legacy stream-copy; also fixes keyframe drift); (2) skip is
LOUD ? provider
433 `metrics["oversize_skipped"]` + segmenter aggregation (chunked + single-file) ?
`logger.error` + NEW thread-safe
434 `backend/reporting/run_signals` channel (incr/pop by video_id; popped unconditionally in
`emit_indexer_report`);
435 (3) report flags it ? `ai_handoff.details.oversize_clips_skipped` + spec.yaml ERROR rule
`oversize_clip_skipped`.
436 Suite 325? **336 green** locally (CI runs fewer: reporting tests importorskip obsreport,
absent there BY DESIGN ?
437 `tests/test_run_signals.py` deliberately has NO importorskip = the CI-visible half of
the channel). **Merge dance
438 (process note):** owner merged #80 mid-session (auto-deleting the remote branch) WHILE
the follow-up test commit was
439 being pushed ? push re-created the branch orphaned; owner rescued it as #81, 39s before
my cherry-picked #82 ?
440 **#82's squash `f01387e` = EMPTY commit** (harmless dup in history). Lesson: re-check
`gh pr view` state before pushing
441 more commits to a PR branch ? owner + sibling sessions work concurrently (this
checkpoint file too: another session

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442     prepended the GEMINI-BASELINE checkpoint mid-edit). ? **OWNER/SIBLING WIP IN TREE
(uncommitted, do NOT clobber):**
443     `backend/providers/gemini.py` ? generate retries 3?6 w/ exponential backoff + jitter
(~3?96s) for the 503 capacity
444     waves. ?? commit/PR the 503-backoff WIP . machine-side 4K re-run (chunks should be ~tens
of MB) . task-17 battery
445     still pending Gemini billing top-up.
446
447     **Checkpoint:** 2026-06-10 (M0 GREENLIT SEQUENCE EXECUTED ? session wrap). Owner
ratified ADR-008 ("split driven by
448     productionization, not isolation") + greenlit M0.4+M0.1. **All 5 steps shipped, every PR
merged on verified-green CI:**
449     - **#73 ADR-008** (DECISIONS.md): training in-repo behind an executable wall; artifact-
bundle boundary; ship-gate
450     product-side; pre-committed split trigger ? `sports-temporal-models` at Gen-2/second-
consumer.
451     - **#74 canonical GT loader** `backend/eval/gt_loader.py::load_gt_intervals` ? closes
the two-loader fork (accepts BOTH
452     `start,end` and `start_time,end_time`); legacy loaders delegate; round-trip contract
test; LOVO 0.626 re-verified.
453     - **#75+#76 featurizer promotion + import wall** : `backend/features/rally_seq.py`
(FEATURE_SCHEMA v1, fps semantics
454     explicit) imported by train AND serve; `training/` skeleton (5 wall rules,
requirements-train.txt); smoke tests:
455     runtime no-training-in-backend.main, static sweep, featurizer purity (torch/cv2
blocked; importorskip numpy on the
456     minimal CI lane ? #76 hotfixed a red lane after #75 merged early; merges now gated on
verified SUCCESS rollout).
457     - **collector #19 ? n=6 OWNED CORPUS IN THE SoR** : all 6 golden matches via `import-
local --normalize` (262 rallies,
458     0 dropped, Proprietary, explicit fps); commercial export now 6 real matches + demo
entry.
459     - **#77 Gen-0 harness** `python -m training.gen0.harness --manifest ? --floor
eval_baselines/heuristic_lovo_n6.json
460     --version 0.1.0` : train?honest LOVO?ship-gate?artifact bundle (gitignored weights.json
+ manifest.json w/ schema pin,
461     calibration, lineage gt_hashes, attestations incl. C3, gate verdict). **First real
bundle built: LOVO 0.626, floor

```

92. assistant (2026-06-13T19:59:24.475Z)

This is the live checkpoint log (newest first). I'll prepend a concise entry for PR #120. Let me insert it right after the header.

93. user (2026-06-13T19:59:35.825Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

94. user (2026-06-13T19:59:42.165Z)

```

1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-14 (ANALYZERS DOC ? FRAMEWORK + paper framing ? **PR #119 OPEN,
owner-gated, do NOT merge** ; branch

```


13 `docs/analyzers-framework` off master `35a50f3`; docs-only). Owner fed two more ideas this session and asked to fold them in

95. user (2026-06-13T19:59:55.324Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

96. assistant (2026-06-13T20:00:14.171Z)

Done. Both changes are implemented, tested, and shipped as a single PR off `master`.

PR: [#120 ? quieter `run.log` + end-of-run rally summary](https://github.com/avidullu/badminton-highlight-ind

Branch `feat/runlog-readability` off `origin/master` @ `35a50f3`; one commit; `git log origin/master..HEAD` shows only mine. Staged explicit paths only ? the sibling session's `docs/`, `artifacts/`, `scratch/` changes are untouched. ****Not merged (owner-gated).****

****Change 1 ? suppress HTTP chatter (config-gated)****

- New central helper backend/utils/logging_setup.py::setup_logging(verbose_http=False) ? raises `httpx`, `httpcore`, `urllib3`, `google\_genai/google.genai`, `google.auth`, `grpc` to `WARNING` (the per-request lines from the Gemini path); `verbose\_http=True` ? `NOTSET` to restore for debugging. `backend.\*` left alone.
- New knob `logging.verbose\_http: false` (backend/config/models.py).
- Wired into run_process_and_stitch.py at import (quiet default) and re-applied after config load.

****Change 2 ? end-of-run rally summary****

- New pure helper backend/tools/export.py::format_rally_summary(segments) -> str, next to `segments\_to\_csv`/`basic\_stats`. Sourced from the produced play_segments so it lines up row-for-row with the rally CSV. Sample:
...

=== Rally summary: 3 rallies, 01:00:05 (3605.4s) total play ===

#	start	end	dur(s)	shots
1	00:00:10	00:00:25	15.4	8
2	00:01:35	01:01:20	3585.0	11
3	01:01:40	01:01:45	5.0	-

...

- Logged as one block at the end of `main()` (and on the all-filtered early-return). HH:MM:SS via a local `\_fmt\_hms` matching the segmenters' `\_fmt\_ts`.

****Verification:**** `python run\_all\_tests.py` ? ****477 passed****; `ruff check .` clean; `mypy backend training` clean. New tests: `test\_logging\_setup.py`, `test\_rally\_summary.py`, plus default/override in `test\_config.py`.

****Two notes:****

- I scoped the wiring to `run\_process\_and\_stitch.py` (the manual-QA "run the tool on a video" path) to stay minimal. The FastAPI server (`backend/main.py`) still has its own `basicConfig` and isn't quieted ? say the word if you want the helper wired there too.
- Heads-up for the record: my first commit-message attempt got mangled (I used PowerShell `@'?'@` here-string syntax inside the ****Bash**** tool, which takes `@` literally). I caught it and amended cleanly via `-F file`; the final message is correct.

Machine-side check is yours (the dev container has no GPU/torch/cv2): run it on a real video and confirm the log is quieter and the summary block reads cleanly against the CSV.

97. user (2026-06-13T20:00:16.676Z)

can you merge the PR as well and ensure the master is clean from your side. I want to resolve this branch collisio

98. assistant (2026-06-13T20:01:43.805Z)

The owner has asked me to merge #120, and this is exactly the kind of good-engineering-fix that [[working-agreement-merging]] authorizes me to self-merge on verified-green CI ? now with an explicit request on top.

Quick note on approach: ultracode is on, but a merge + working-tree cleanup on a ****shared checkout**** is a serial, stateful git operation. Fanning out parallel agents here would race on ``.git/index`/`HEAD`` and risk clobbering the concurrent session's uncommitted WIP ? the opposite of what's wanted. So I'll do this directly and carefully, with thorough read-only verification at each step. Let me first establish the exact state (read-only).

99. user (2026-06-13T20:01:48.344Z)

```
==== PR #120 state ====
{"baseRefName":"master","headRefName":"feat/runlog-readability","isCrossRepository":false,"merge
StateStatus":"CLEAN","mergeable":"MERGEABLE","number":120,"state":"OPEN"}
```

```
==== PR #120 checks ====
Full test suite (offline, mocked)      pass    1m31s    https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27477491781/job/81219327487
Mypy (lenient baseline) pass          34s          https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27477491781/job/81219327498
Ruff lint                               pass     9s          https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27477491781/job/81219327531
Syntax sweep + import smoke (no GPU stack) pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27477491781/job/81219327484
```

```
==== local branch ====
feat/runlog-readability
```

```
==== git status (porcelain) ====
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
?? scratch/arial.ttf
?? scratch/debug_wsl.py
?? scratch/keep_awake.ps1
?? scratch/p3_smoke.py
?? scratch/probe_cv2_count.py
?? scratch/probe_gemini.py
?? scratch/repro_drawtext_font.py
?? scratch/repro_movie_overlay.py
?? scratch/run_gx010125.py
?? scratch/stitch_min5.py
?? scratch/trim_video.py
```

```
==== staged diff stat ====
```

100. assistant (2026-06-13T20:02:15.021Z)

CI is fully green (all 4 checks pass), ``mergeable: CLEAN``, and the working tree now has ****no tracked-file modifications**** ? only the intentional untracked carryovers (``scratch/``, ``artifacts/``). The sibling's previously-staged ``docs/ANALYZERS_AND_RECONCILERS.md`` change is no longer in the tree, so there's no WIP of theirs for me to preserve in the index. Let me re-fetch and confirm ``origin/master`` hasn't moved under me before the merge.

101. user (2026-06-13T20:02:19.929Z)

```
==== origin/master tip ====
35a50f3 docs(readme): index QUALITY_ITERATIONS + mark EXPERIMENT_HARNESS P1 SHIPPED (#118)
==== my branch tip ====
```

```
3af7d1a feat(logging): quieter run.log ? suppress HTTP chatter + end-of-run rally summary
==== commits I'm adding (origin/master..HEAD) ====
3af7d1a feat(logging): quieter run.log ? suppress HTTP chatter + end-of-run rally summary
==== is my branch a clean fast-forward onto origin/master? (master..mybranch should be just my 1
commit; origin/master..nothing-behind) ====
behind origin/master by:
0
```

102. assistant (2026-06-13T20:02:37.127Z)

Clean linear state confirmed: `origin/master` unmoved at `35a50f3`, my branch is exactly
`origin/master` + my single commit `3af7d1a`, 0 behind. CI green, `mergeable: CLEAN`. Squash-
merging now (repo convention is squash), deleting the branch on both sides.

103. user (2026-06-13T20:02:43.094Z)

```
From https://github.com/avidullu/badminton-highlight-indexer
* branch          master      -> FETCH_HEAD
  35a50f3..6076eef master      -> origin/master
Updating 35a50f3..6076eef
Fast-forward
 backend/config/models.py      | 10 ++++++
 backend/tools/export.py        | 59 ++++++++++++++++++++++++++++++++++++++
 backend/utils/logging_setup.py | 56 ++++++++++++++++++++++++++++++++++++++
 run_process_and_stitch.py      | 25 ++++++-----
 tests/test_config.py           | 15 ++++++++
 tests/test_logging_setup.py    | 44 ++++++++++++++++++++++++++++++++++++
 tests/test_rally_summary.py    | 56 ++++++++++++++++++++++++++++++++++++++
 7 files changed, 259 insertions(+), 6 deletions(-)
 create mode 100644 backend/utils/logging_setup.py
 create mode 100644 tests/test_logging_setup.py
 create mode 100644 tests/test_rally_summary.py
```

104. assistant (2026-06-13T20:02:48.811Z)

Merge succeeded ? master advanced `35a50f3..6076eef` and fast-forwarded locally. Let me verify
the complete final state: branch, sync, branch deletion (both sides), PR status, and working
tree.

105. user (2026-06-13T20:02:54.127Z)

```
==== current branch ====
master

==== local master vs origin/master ====
6076eef feat(logging): quieter run.log ? suppress HTTP chatter + end-of-run rally summary (#120)
6076eef feat(logging): quieter run.log ? suppress HTTP chatter + end-of-run rally summary (#120)

==== HEAD ahead/behind origin/master (should be 0 0) ====
ahead:
0
behind:
0

==== local branches (feat/runlog-readability should be GONE) ====
ci/ruff-lint-lane
docs/analyzers-and-reconcilers
docs/analyzers-framework
docs/pmf-field-signals
docs/storage-sharing-model
docs/video-locality-model
fix/stitching-padding-config
* master
```

==== remote branch gone? (grep should be empty) ====

origin/feat/runlog-readability

==== PR #120 state ====

```
{"mergeCommit": "6076eeff1f6c70a6ad49af7100a704b7456b694c", "mergedAt": "2026-06-13T20:02:39Z", "state": "MERGED"}
```

==== working tree (porcelain) ====

```
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
?? scratch/arial.ttf
?? scratch/debug_wsl.py
?? scratch/keep_awake.ps1
?? scratch/p3_smoke.py
?? scratch/probe_cv2_count.py
?? scratch/probe_gemini.py
?? scratch/repro_drawtext_font.py
?? scratch/repro_movie_overlay.py
?? scratch/run_gx010125.py
?? scratch/stitch_min5.py
?? scratch/trim_video.py
```

106. assistant (2026-06-13T20:03:26.509Z)

Core objective achieved: I'm on `master`, local `master == origin/master == 6076eef` (the `(#120)` squash commit), my local feature branch is deleted, PR #120 is **MERGED**, and the working tree has only the intentional untracked carryovers ? no tracked changes, no leftover of mine.

Two cleanup items remain: a stale `origin/feat/runlog-readability` remote-tracking ref (the remote branch was deleted by the merge, but my local ref needs pruning), and a pile of **other sessions**' stale local branches. Let me prune the stale ref and map every local branch to its PR state before touching anything ? I won't delete branches tied to open PRs.

107. user (2026-06-13T20:03:37.668Z)

==== does the remote branch still exist on GitHub? (expect 404) ====

```
{"message": "Branch not found", "documentation_url": "https://docs.github.com/rest/branches/branches#get-a-branch", "status": "404"}gh: Branch not found (HTTP 404)
```

==== prune stale remote-tracking refs ====

Pruning origin

URL: <https://github.com/avidullu/badminton-highlight-indexer.git>

```
* [pruned] origin/docs/readme-index-sync
* [pruned] origin/feat/experiment-harness
* [pruned] origin/feat/person-cue-extraction
* [pruned] origin/feat/runlog-readability
```

==== remote runlog ref after prune (expect empty) ====

(gone ? clean)

==== map each LOCAL branch -> PR state ====

ci/ruff-lint-lane	PR #92 MERGED: ci: ruff lint lane + fix all violations;
remove never-compiled flyer stub	
docs/analyzers-and-reconcilers	PR #116 CLOSED: docs: two rally-detection extensibility seams (injectable analyzers + reconciliation pass)
docs/analyzers-framework	PR #119 OPEN: docs: extend analyzers/reconcilers into a multi-scope signal-fusion framework (warm-up + shuttle-state)
docs/pmf-field-signals	PR #104 MERGED: docs(pmf): C13 field kit ? recruiting

flyer, visit playbook, answer sheet, pre-registered signals
docs/storage-sharing-model PR #111 MERGED: docs(strategy): storage-priced sharing
mental model (DRAFT, co-design)
docs/video-locality-model PR #106 MERGED: docs(platform): video-locality mental
model ? the 10GB question
fix/stitching-padding-config PR #102 MERGED: fix(api): thread config stitching
paddings into live VideoCompiler sites

108. assistant (2026-06-13T20:04:25.822Z)

Clear mapping. The remote branch is confirmed deleted on GitHub (404), and I pruned 4 stale remote-tracking refs (including mine). Now the local branches:

```
| Local branch | PR | State | Action |
|---|---|---|---|
| `ci/ruff-lint-lane` | #92 | MERGED | delete |
| `docs/analyzers-and-reconcilers` | #116 | CLOSED (redundant; content on master via #117) | delete |
| `docs/pmf-field-signals` | #104 | MERGED | delete |
| `docs/storage-sharing-model` | #111 | MERGED | delete |
| `docs/video-locality-model` | #106 | MERGED | delete |
| `fix/stitching-padding-config` | #102 | MERGED | delete |
| `docs/analyzers-framework` | **#119 OPEN** | **active, owner-gated** | **KEEP** |
```

This matches your established protocol (prune local branches that map to merged/closed PRs; keep `master` + open-PR branches) and directly resolves the collision clutter. Branch deletion is local-only and reflog-recoverable. Deleting the 6 stale ones, keeping `master` + the live #119 branch.

109. user (2026-06-13T20:04:28.108Z)

```
==== safety: confirm on master ====
master
```

```
==== delete the 6 stale local branches (merged/closed PRs) ====
Deleted branch ci/ruff-lint-lane (was 75659d1).
Deleted branch docs/analyzers-and-reconcilers (was 0f673ff).
Deleted branch docs/pmf-field-signals (was 9c84fea).
Deleted branch docs/storage-sharing-model (was adbe579).
Deleted branch docs/video-locality-model (was 85abe26).
Deleted branch fix/stitching-padding-config (was 30e4899).
```

```
==== remaining local branches (expect: docs/analyzers-framework + master) ====
docs/analyzers-framework
* master
```

110. assistant (2026-06-13T20:04:39.037Z)

All 6 stale branches deleted; only `master` (current) and `docs/analyzers-framework` (#119 OPEN) remain locally. Final verification of the whole clean state:

111. user (2026-06-13T20:04:43.529Z)

```
==== on branch ====
master
```

```
==== master tip (should be #120 squash) + last 3 ====
6076eef feat(logging): quieter run.log ? suppress HTTP chatter + end-of-run rally summary (#120)
35a50f3 docs(readme): index QUALITY_ITERATIONS + mark EXPERIMENT_HARNESS P1 SHIPPED (#118)
cbf2236 feat(segmenters): FusionSegmenter ? shuttle + net-crossing + person-cue fusion (P2, default-OFF) (#117)
```

```
==== sync with origin/master (0 ahead / 0 behind) ====
ahead:
```

```

0
behind:
0

==== local branches ====
  docs/analyzers-framework
* master
==== remote branches still on origin ====
  origin/HEAD -> origin/master
  origin/ci/ruff-lint-lane
  origin/claude/docs-handoff-consistency
  origin/claude/highlight-watermark
  origin/claude/indexer-doc-review-XVPTh
  origin/claude/platform-architecture-research
  origin/claude/watermark-bundled-font
  origin/docs/analyzers-and-reconcilers
  origin/docs/analyzers-framework
  origin/docs/next-steps-multisport
  origin/docs/pmf-field-signals
  origin/docs/storage-sharing-model
  origin/docs/video-locality-model
  origin/feat/async-job-model
  origin/feat/fusion-segmenter
  origin/feat/khelacharya-shot-intelligence
  origin/feat/owned-model-study-and-prototype
  origin/fix/stitching-padding-config
  origin/master

==== working tree (only untracked carryovers expected) ====
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
?? scratch/arial.ttf
?? scratch/debug_wsl.py
?? scratch/keep_awake.ps1
?? scratch/p3_smoke.py
?? scratch/probe_cv2_count.py
?? scratch/probe_gemini.py
?? scratch/repro_drawtext_font.py
?? scratch/repro_movie_overlay.py
?? scratch/run_gx010125.py
?? scratch/stitch_min5.py
?? scratch/trim_video.py

==== merged code present on master? ====
  backend/utils/logging_setup.py ? on master
  format_rally_summary defs on master HEAD: 1

```

112. assistant (2026-06-13T20:05:16.351Z)

My side is fully clean: on `master` at the `(#120)` squash commit, 0 ahead/0 behind `origin/master`, only `master` + the live `docs/analyzers-framework` (#119) remain locally, working tree has nothing but the intentional untracked carryovers, and the merged code is confirmed on master.

One thing surfaced that's **your call**, not mine to assume: the **remote** (`origin`) still carries 16 stale branches. Most map to already-merged/closed PRs, but a few are deliberately kept or still active. Deleting remote branches is outward-facing on the shared repo, so let me map them precisely before touching anything.

113. user (2026-06-13T20:05:32.226Z)

branch PR/state	
ci/ruff-lint-lane	PR #92 MERGED
claude/docs-handoff-consistency	PR #59 MERGED
claude/highlight-watermark	PR #101 MERGED
claude/indexer-doc-review-XVPTH	PR #113 MERGED
claude/platform-architecture-research	PR #58 MERGED
claude/watermark-bundled-font	PR #103 MERGED
docs/analyzers-and-reconcilers	PR #116 CLOSED
docs/analyzers-framework	PR #119 OPEN
docs/next-steps-multisport	PR #62 MERGED
docs/pmf-field-signals	PR #104 MERGED
docs/storage-sharing-model	PR #111 MERGED
docs/video-locality-model	PR #106 MERGED
feat/async-job-model	PR #87 MERGED
feat/fusion-segmenter	PR #117 MERGED
feat/khelacharya-shot-intelligence	PR #11 CLOSED
feat/owned-model-study-and-prototype	PR #61 MERGED
fix/stitching-padding-config	PR #102 MERGED

114. assistant (2026-06-13T20:06:13.492Z)

Full remote mapping is clear. Before I update memory and report, here's the situation on the **remote**:

- **Safe to prune (15):** 14 branches whose PRs are MERGED (`ci/ruff-lint-lane` #92`, `claude/docs-handoff-consistency` #59`, `claude/highlight-watermark` #101`, `claude/indexer-doc-review-XVPTH` #113`, `claude/platform-architecture-research` #58`, `claude/watermark-bundled-font` #103`, `docs/next-steps-multisport` #62`, `docs/pmf-field-signals` #104`, `docs/storage-sharing-model` #111`, `docs/video-locality-model` #106`, `feat/async-job-model` #87`, `feat/fusion-segmenter` #117`, `feat/owned-model-study-and-prototype` #61`, `fix/stitching-padding-config` #102`) + `docs/analyzers-and-reconcilers` #116` (CLOSED, content already on master via #117).
- **Must keep:** `docs/analyzers-framework` (#119 OPEN)`, `feat/khelacharya-shot-intelligence` (#11 closed but deliberately retained` as the home for the sibling shot-intelligence layer ? per project memory), and `master``.

Your local checkout is already fully clean and the collision is resolved. Pruning those 15 **remote** branches is outward-facing on the shared repo and includes a deliberately-kept branch in the neighborhood, so I want your explicit go-ahead on scope rather than assume it.

115. user (2026-06-13T20:06:20.130Z)

The user did not answer the questions.

116. assistant (2026-06-13T20:06:26.903Z)

I'll leave the remote branches untouched without an explicit go-ahead ? the merge and the local cleanup (the actual collision) are done. Let me refresh the memory checkpoint to reflect the merged state, since a PR-merged is a checkpoint event. First I'll re-read the entry I wrote earlier to edit it precisely.

117. user (2026-06-13T20:06:28.237Z)

```

1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5        node_type: memory
6        type: project
7        originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
```

```

12      **Checkpoint:** 2026-06-14 (RUN.LOG READABILITY for manual QA ? **PR #120 OPEN, owner-
gated, do NOT merge**); branch
13      `feat/runlog-readability` off master `35a50f3`; CODE, minimal/config-gated). Owner's
manual-QA loop = run the tool on a video ?
14      eyeball the rally CSV against `run.log`. Two readability changes: (1) **suppress HTTP
chatter** ? new central helper
15      `backend/utils/logging_setup.py::setup_logging(verbose_http=False)` raises
`httpx/httpcore/urllib3/google_genai/google_genai/
16      google.auth/grpc` to WARNING (NOTSET to restore), gated by NEW knob
`logging.verbose_http: false` (`LoggingConfig` on
17      `AppConfig`); our `backend.*` loggers untouched; wired into `run_process_and_stitch.py`
(at import + re-applied
18      post-config-load). (2) **end-of-run rally summary** ? new pure helper
`backend/tools/export.py::format_rally_summary` (next to
19      `segments_to_csv`/`basic_stats`): header (total rallies + total play HH:MM:SS) + table
`# start end dur(s) shot_count` from the
20      produced play_segments so log?CSV line up row-for-row; handles metadata dict-or-JSON-
string, missing shot_count?`-`; HH:MM:SS
21      via local `_fmt_hms` matching the segmenters' `_fmt_ts`; logged as ONE block at end of
`main()` (also on the all-filtered
22      early-return). Tests: `test_logging_setup.py`, `test_rally_summary.py`, + `verbose_http`
default/override in `test_config.py`.
23      Suite **477 green**, ruff (`. `) + mypy (`backend training`) clean. ? Commit-msg first
attempt mangled (PowerShell `@'?'@`
24      here-string in the *Bash* tool ? literal `@`; amended via `-F file`) ? Bash tool ?
PowerShell. ?? NEXT: owner reviews/merges
25      #120 (machine-side: run on a real video to eyeball the quieter log + summary block; dev
container can't run the GPU pipeline).
26
27      **Checkpoint:** 2026-06-14 (ANALYZERS DOC ? FRAMEWORK + paper framing ? **PR #119 OPEN,
owner-gated, do NOT merge**); branch
28      `docs/analyzers-framework` off master `35a50f3`; docs-only). Owner fed two more ideas
this session and asked to fold them in

```

118. assistant (2026-06-13T20:06:34.633Z)

Now I'll update the checkpoint to reflect the merge + cleanup:

119. user (2026-06-13T20:06:38.611Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

120. user (2026-06-13T20:06:49.225Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

121. assistant (2026-06-13T20:07:05.203Z)

Done. Here's the final state.

? PR #120 merged

- All 4 CI checks were green and `mergeable: CLEAN`, so I squash-merged (repo convention). Master advanced `35a50f3` ? 6076eef` ? commit [`(#120)`](https://github.com/avidullu/badminton-highlight-indexer/commit/6076eef). PR is \*\*MERGED\*\*; the `feat/runlog-readability` branch was deleted on **both** remote and local. This was within [[working-agreement-merging]] (good-engineering-fix, green CI) plus your explicit ask.

? Master clean from my side / collision resolved

- On `master`, `local master == origin/master == 6076eef`, **0 ahead / 0 behind**.

- My feature branch is gone; the merged code is confirmed live on master (``backend/utils/logging_setup.py`` present, ``format_rally_summary`` on HEAD).
- Working tree carries **only** the intentional untracked carryovers (``scratch/``, ``artifacts/``) ? no tracked changes, nothing of mine left behind. The sibling's previously-staged ``docs/ANALYZERS_AND_RECONCILERS.md`` edit had already cleared from the index before I merged, so there was no WIP of theirs to disturb.
- **Pruned 6 stale local branches** (all mapped to merged/closed PRs: #92, #104, #111, #106, #102, and #116-redundant) + 4 stale remote-tracking refs. **Kept** ``docs/analyzers-framework`` (#119, still OPEN). Local now shows just ``master`` + the one live branch.

? Remote branches ? left untouched (your call)

The shared ``origin`` still has 15 prunable branches (14 merged-PR + the redundant #116). I asked before touching them because remote deletion is outward-facing and the set neighbors ``feat/khelacharya-shot-intelligence`` (#11) ? which project memory says is **deliberately retained** as the sibling shot-intelligence layer, so it must not be swept. You didn't answer, so I left the remote as-is. The offer stands ? just say "prune the remote stale branches" and I'll delete those 15 while keeping #119, the khelacharya branch, and ``master``.

The memory checkpoint is updated to reflect the merge + cleanup. Remaining for you: the machine-side eyeball (run the tool on a real video to confirm the quieter log and the summary block read well against the CSV) ? the dev container can't run the GPU pipeline.